

Malware Patching Strategies in Edge Intelligence IoT Systems: A Differential Games Approach With Spiking Neural Networks

Yizhou Shen , Carlton Shepherd , Chuadhry Mujeeb Ahmed , Shigen Shen , *Senior Member, IEEE*,
and Shui Yu , *Fellow, IEEE*

Abstract—Malware poses a significant threat to Internet of Things (IoT) systems, leading to unauthorized access and data breaches. Evolving malware continues to pose severe threats to IoT systems, necessitating defense mechanisms capable of adaptive learning and real-time strategic decision-making. To this end, we present a novel IoT malware propagation and patching model based on differential games and spiking neural networks in edge intelligence (EI)-enabled IoT systems. Specifically, built upon optimal control theory, we reveal the process of dynamic evolution between infected IoT devices and associated edge nodes based on differential games under malware propagation in EI-enabled IoT systems. The dynamic state transitions are described by differential equations, followed by introducing attack/defense intention for theoretically attaining the optimal IoT malware patching strategies. Moreover, we design a novel IoT malware propagation-patching approach named differential games-based deep spiking Q-network (DGDSQ) for practical patch optimization decisions in EI-enabled IoT systems. Additionally, we conduct experimental simulations to comprehend that DGDSQ-assisted edge nodes are more effective than double deep Q-network and dueling double deep Q-network against IoT malware propagation.

Index Terms—Edge intelligence, Internet of Things, differential games, spiking neural networks, deep reinforcement learning, malware propagation.

I. INTRODUCTION

MALWARE has been a major issue for the Internet of Things (IoT) for a long time [1], [2]. Once malware invades IoT devices, it can rapidly propagate through interconnected devices and compromise the stability of the entire IoT system [3], [4], [5]. Several real-world incidents highlight the severity of this problem. For example, the WannaCry ransomware outbreak in 2017 caused widespread disruption across

multiple National Health Service (NHS) trusts [6], while the Synnovis cyber attack in 2024 disrupted pathology services in several London hospitals, leading to cancelled appointments and delayed tests [7]. IoT-specific malware further amplifies this threat due to the large-scale and heterogeneity of IoT devices. The Mirai botnet infected thousands of IoT devices such as cameras and routers to launch massive distributed denial-of-service (DDoS) attacks, while BrickerBot permanently disabled vulnerable IoT devices by corrupting their firmware [8]. These examples show that once malware spreads through vulnerable IoT devices, it can rapidly escalate into large-scale system disruption. Due to limited resources in IoT devices, it is often necessary to rely on edge nodes to complete the corresponding computational tasks. However, it is difficult to detect all infected IoT devices and maintain normal operations [9]. Besides, while other works based on cryptographic methods [10], [11] and differential privacy [12], [13] offer data protection, they do not address the dynamic and adversarial nature of malware propagation. It is notable that deep neural networks (DNN) can extract features from massive data and solve decision-making problems by combining deep reinforcement learning (DRL) through interactive learning with the environment [14], [15], [16]. Thus, DRL-based edge nodes can be utilized to sense the status of IoT malware and predict malware propagation behavior of infected IoT devices, followed by deriving the optimal malware patching action, which forms IoT malware patching strategy, so as to achieve intelligent detection and recovery [17], [18]. To this end, with the assistance of edge computing and DRL, we aim to enable such edge nodes to automatically explore optimal learning strategies under IoT malware propagation in edge intelligence (EI)-enabled IoT systems.

We model the system as a competitive environment where infected IoT devices and associated edge nodes act as opposing players, both of which pursue their own maximal interests. Under this circumstance, game theory [19] has been used for exploring optimization strategies with the consideration of the predicted and actual behavior of players. Integrating optimal control into game theory, differential games [20] not only makes it possible to obtain the optimal control strategy for a dynamic system but also maximizes the rewards in the multi-player multistrategy decision-making process [21], [22]. In a differential game, differential equations are applied to describe player actions based on state transitions. Each player strives

Received 4 October 2025; revised 25 May 2026; accepted 26 May 2026. Date of publication 29 May 2026; date of current version 1 September 2026. The work of Carlton Shepherd was supported by U.K. EPSRC ‘Chameleon’ project under Grant EP/Y030168/1. (Corresponding author: Carlton Shepherd.)

Yizhou Shen and Chuadhry Mujeeb Ahmed are with the School of Computing, Newcastle University, NE1 7RU Newcastle upon Tyne, U.K. (e-mail: y.shen34@newcastle.ac.uk; mujeeb.ahmed@newcastle.ac.uk).

Carlton Shepherd is with the Department of Computer Science, Durham University, DH1 3LE Durham, U.K. (e-mail: carlton.g.shepherd@durham.ac.uk).

Shigen Shen is with the School of Information Engineering, Huzhou Normal University, Huzhou 313000, China (e-mail: shigens@zjhu.edu.cn).

Shui Yu is with the School of Computer Science, University of Technology Sydney, Ultimo, NSW 2007, Australia (e-mail: shui.yu@uts.edu.au).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TDSC.2026.3698145>, provided by the authors.

Digital Object Identifier 10.1109/TDSC.2026.3698145

to optimize their independent and conflicting strategies until reaching Nash equilibrium where no players have a motivation to deviate from their currently chosen strategies. Compared with commonly used reinforcement learning or traditional control approaches, differential games can provide several advantages for modeling malware propagation in IoT systems. Reinforcement learning methods typically rely on large amounts of interaction data and training time to learn effective strategies. However, malware defense scenarios often require rapid decision-making and theoretical guarantees. Differential games enable analytical modeling of dynamic interaction between attackers and defenders, followed by providing equilibrium-based optimal strategy. Different from other game types such as stochastic games [23], [24], differential games explicitly capture the continuous-time evolution of system states and strategies. This makes them well suited for modeling malware propagation processes, which evolve dynamically with device states and defense actions. Therefore, differential games offer an effective tool for analyzing competition and cooperation in multiplayer scenarios, particularly for controlling uncertain dynamic systems in IoT environments [25], [26].

However, deploying DNN-based DRL in resource-constrained IoT environments faces significant limitations. Specifically, fully-connected (FC) hidden layers in DNN lead to a substantial expansion in the number of parameters, increasing computational overhead. Moreover, issues such as local optima and gradient vanishing become more prominent as the number of hidden layers increases, affecting the effective training of models for complex security tasks. To address these limitations, we develop spiking neural network (SNN)-based DRL methods. As a new generation of biologically inspired models, SNN-based DRL offers lower computational complexity and reduces energy-related operations. This makes them ideally suited for enabling intelligent malware patching and device recovery in IoT systems [27], [28]. Furthermore, the event-driven nature of SNNs allows strategy selection at edge nodes to more closely resemble efficient brain-like decision-making [29], [30]. Therefore, edge nodes assisted by SNN-based DRL can autonomously learn optimal malware patching strategies through continuous learning, which enhances autonomous action prediction and decision-making.

Based on the above, several noteworthy research questions appear as follows:

- 1) Can differential games properly describe IoT malware dynamic evolution and predict possible behavior for infected IoT devices and associated edge nodes in EI-enabled IoT systems?
- 2) Can SNN successfully replace DNN and combine it with DRL algorithms to realize more advanced intelligent IoT malware propagation-patching decision-making in EI-enabled IoT systems?
- 3) Which potential parameters will influence IoT malware patching strategy selection in EI-enabled IoT systems?

To address these questions, we propose a differential game-based IoT malware propagation-patching model to recover infected IoT devices in EI-enabled IoT systems. We first analyze state transitions for IoT devices and edge nodes under IoT

malware propagation, followed by revealing dynamic evolution for both players based on differential equations. It is notable that attack/defense intentions are additionally introduced to achieve a more realistic representation of the IoT malware propagation-patching process for both players. We then implement an SNN-based DRL algorithm to practically solve the optimization problem and obtain the optimal patching strategy for defending IoT malware propagation in EI-enabled IoT systems. Here, traditional DNNs are replaced by novel SNNs to make edge nodes more closely simulate human decision-making processes, which helps improve action prediction and optimize IoT malware patching strategies. Furthermore, we conduct experimental simulations to explore the optimal patching probability and infection rate under malware propagation, as well as demonstrate the superiority of the proposed method in EI-enabled IoT systems. To the best of our knowledge, this is the first work to mitigate IoT malware propagation by combining differential games and SNN-based DRL algorithm in EI-enabled IoT systems.

The main contributions are summarized as follows:

- 1) We model differential games between IoT devices and edge nodes to characterize the dynamic interaction between malware propagation and defense in EI-enabled IoT systems. Different from existing IoT security models that often rely on static or discrete-time formulations, the proposed differential game captures the continuous-time evolution of attack and defense strategies using differential equations. This enables a more realistic representation of malware propagation dynamics and the adaptive patching process.
- 2) We integrate SNN into a DRL framework to solve the proposed malware propagation-patching game. Unlike conventional DRL approaches that employ standard artificial neural networks, the SNN-based DRL model exploits the temporal dynamics and energy-aware characteristics of spiking neurons to learn control strategies for malware patching. This provides a novel learning-based solution for implementing adaptive defense mechanisms in EI-enabled IoT environments.
- 3) We analyze the influence of patching probability and infection rate on state transition rewards, probabilities of IoT devices being in two specific states, and the frequency of attack/defense intentions, respectively. In addition, we compare the performance of the basic DRL algorithms and SNN-based DRL and demonstrate the advantages of the proposed SNN-based DRL approach.

The rest of the paper is arranged as follows. In Section II, we review the security and application of EI-enabled IoT systems, the development of security-oriented differential games-based models, and the implementation of SNN-assisted DRL approaches. In Section III, we form a differential games-based model to disclose competitive interactions and dynamic transitions between IoT devices and edge nodes in EI-enabled IoT systems with IoT malware propagation. In Section IV, we raise an SNN-based DRL algorithm to practically control IoT malware propagation and acquire the optimal EI-enabled IoT malware patching strategy. In Section V, we numerically evaluate the

effect of parameter-settings on optimal decision-making, and compare the performance among three DRL algorithms, which is followed by the conclusion and future work in Section VI.

II. RELATED WORK

As a new generation technology that enables real-time data processing and intelligent decision-making, EI-related studies, including EI security and applications, continue to emerge. Akter et al. advised a federated edge aggregator framework with artificial noise for preserving the privacy of EI-based federated learning in IoT smart healthcare systems [31]. An et al. proposed an EI-assisted anomaly detection approach for hypertext transfer protocol (HTTP) in IoT systems, in which the unusual HTTP traffic can be quickly and precisely discovered based on clustering and classification [32]. Makkar et al. investigated an AI-supported web spam detection framework based on edge computing in IoT systems, in which a cognitive intrusion security model is established to guarantee the reliability of search engine results and resist web spam [33]. Xu et al. delved into a smart contract-driven EI-assisted scheme based on blockchain and federated learning to solve the trust issues among unfamiliar IoT devices and security issues among centralized aggregators [34]. Wanode et al. concluded a machine intelligence-aided IoT fingerprinting method on edge infrastructure to reveal the optimal feature set and protect it from various security attacks [35].

With the combination of optimal control and game theory, differential games have been widely implemented in optimal control processes controlled by two parties with opposing interests. Yang et al. put forward a differential game-based repair model for advanced persistent threat, which optimally allocates available repair resources to prevent potential loss of user-sensitive data [36]. Huang and Zhu suggested a differential game-based decentralized virus-resistant weight adaptation framework to mitigate virus propagation across the network, in which the virus propagation process is captured by an epidemic model [25]. Zhang and Xu came up with a differential game-based device-to-device (D2D) offloading malware restrain architecture on heterogeneous networks to gain preventive defense strategies [37]. Wu et al. recommended a differential game-based attack-defense strategy analysis to explore the network resource competition and strategy interaction between attackers and defenders as well as the real-time evolution of system security states in IoT systems [21]. Considering the dynamics and continuity of network interaction, Gao et al. applied differential games to research the real-time decision-making between multi-attacker and multi-defender in massive machine-type communications systems, which derives the evolution of the optimal attack-defense strategy [38]. Ghosh and Konstantinou developed a differential game-based cyber-physical model for load frequency control to resist false data injection attacks, which ensures the normal operation of the control center and generates proper control signals [39].

Owing to the unavailability of exact game parameters such as rewards in practice [40], we employ DRL trained via SNN to approximate the optimal strategy. Liu et al. propounded a deep spiking Q-network to achieve direct training based

on leaky integrate-and-fire (LIF) neurons, which solves the problem that the discrete spikes output by SNN are unable to represent the highly similar Q-values [41]. The directly trained SNN provides a novel method to achieve high performance in Atari games and refrains from the limitations of conversion approaches. Oikonomou et al. raised a hybrid deep deterministic policy gradient (DDPG) consisting of SNN-based actor networks and DNN-based critic networks for bioinspired robotic arm target reaching, which enables the spiking agent to carry out autonomous exploration and improves the robot learning capability [42]. Similarly, Tang et al. modeled another DDPG framework with SNN actor networks and DNN critic networks to achieve energy-efficient mapless navigation, where the two networks generate reliable control strategies through cooperative information exchange [43]. Tan et al. designed an SNN-based DRL, in which a robust conversion approach was developed to reduce the error caused by converting deep Q-network (DQN) parameters from DNN to SNN [44]. Chevtchenko and Ludermitr constructed an SNN-based reinforcement learning method with sparse and delayed rewards, which not only addresses the challenge of input dimension scalability but also achieves high stability with low memory usage [45]. Saravanan et al. explored the effectiveness of SNN-based DRLs in single- and multi-agent environments, in which SNN-based DRLs require less data to attain the appropriate strategies with less time, demonstrating the superior performance of SNN-based models [46].

Based on the above analyses, we focus on a differential game-based SNN-assisted IoT malware propagation-patching model, as the first work, to understand the dynamic malware propagation-patching process in EI-enabled IoT systems, which is followed by developing a novel SNN-based DRL algorithm to gain the optimal IoT malware patching strategy. Specifically, we integrate DRL into EI-enabled IoT systems to enable edge nodes to automatically detect and patch various IoT malware, thus achieving self-adaptive learning and intelligent decision-making. Additionally, we incorporate attack/defense intentions into the dynamic state transitions governed by differential equations to more accurately represent the practical process of IoT malware propagation and patching. Finally, we respectively seek the optimal parameter-settings of patching probability and infection rate, as well as validate the superiority of the proposed algorithm.

Table I presents the differences between our work and other SNN-based approaches in terms of scenario, solution, advantage, and limitation.

III. MODELS

A. System Model

Owing to the networked nature of IoT systems and the vulnerabilities of IoT devices, we consider that *susceptible* (S), *latent* (L), *infected* (I), *recovered* (R), and *dead* (D) are five states of IoT devices during malware propagation in EI-enabled IoT systems. To be specific, state S represents IoT devices that are susceptible to infection by IoT malware. State L represents IoT devices that are infected by dormant IoT malware, meaning they

TABLE I
COMPARATIVE TABLE EXPLAINING DIFFERENCES BETWEEN OUR WORK AND OTHER SNN-BASED APPROACHES

Paper	Scenario	Solution	Advantage	Limitation
Liu et al. [42]	Atari games	Directly trained DSQN based on LIF neurons	Solve the issue of representing Q-values with discrete spikes	Incomplete theoretical analysis of LIF neuron dynamics
Oikonomou et al. [43]	Robotic arm target reaching	Hybrid DDPG with an SNN-based actor and a DNN-based critic	Enable autonomous exploration and improve robot learning capability with a bio-inspired approach	Limited to target reaching without object manipulation capabilities
Tang et al. [44]	Energy-efficient mapless navigation	DDPG framework with SNN actor and DNN critic, using cooperative information exchange	Generate reliable control strategies and achieve energy-efficient navigation	Complex co-learning process using SNN and DNN
Tan et al. [45]	General DRL tasks	SNN-based DRL using a robust conversion method to transfer parameters from a trained DQN	Reduce errors associated with converting DNN parameters to SNN, leading to more accurate SNN models	Occasional selection of non-optimal actions by SNN agents
Chevtchenko and Ludermit [46]	Environments with sparse and delayed rewards	Multilayer spiking neural network for reinforcement learning	Address input dimension scalability and achieve high stability with low memory usage	Limited scalability to large action spaces
Saravanan et al. [47]	Single- and multi-agent environments	Exploration of SNN-based DRLs in various agent settings	Require less data and time to learn effective strategies	Limited improvement on multi-agent environments
Our paper	IoT malware propagation	Differential game-based SNN-assisted DRL framework	Achieve SNN-based DRL for automated malware patching	Reliance on simulated transition probabilities rather than real-world empirical data

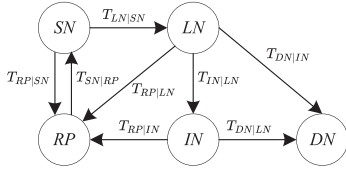


Fig. 1. State transitions of IoT devices under malware propagation in EI-enabled IoT systems.

do not actively propagate the malware. On the contrary, state *I* represents IoT devices that have been infected by active IoT malware and they can contaminate other IoT devices. Specifically, infected IoT devices can further propagate malware to susceptible devices through network communications. State *R* represents infected IoT devices that have been recovered by installing security patches to restrain malware propagation. State *D* represents infected IoT devices that have been rendered inoperable.

To defend against malware propagation through IoT devices, we additionally consider that *patched* (*P*) and *non-patched* (*N*) are two actions of edge nodes in EI-enabled IoT systems. Action *P* represents edge nodes successfully installing patches and action *N* represents edge nodes opting not to install patches.

Based on the analyses above, we define five joint states: *SN*, *LN*, *IN*, *RP* and *DN*. Herein, state *SN* represents that IoT devices are susceptible and not patched, state *LN* represents that IoT devices are in latency and not patched, state *IN* represents that IoT devices are infectious but not patched, state *RP* represents that infected IoT devices have been patched and recovered, and state *DN* represents that infected IoT devices are dead and were not patched. Thus, the joint state transition of IoT devices belonging to the EI-enabled IoT system can be constructed as shown in Fig. 1. It is worth noting that this five-state transition model is consistent with empirical analyses of IoT malware lifecycles [47], [48]. Similar state transitions have been extensively adopted in epidemic models such as SIR [49], [50], where they effectively capture IoT malware infection and containment. By extending classical epidemic frameworks, we adopt a five-state model to incorporate a latent probing stage before full compromise. Devices in this stage may either be successfully patched or become permanently disabled, thereby capturing the essential dynamics of malware propagation and defense in EI-enabled IoT systems. In summary, the proposed state transition model is a comprehensive epidemic framework that describes

TABLE II
PARAMETERS FOR MALWARE PROPAGATION AND PATCHING

Symbol	Definition
ϱ_n^i	Probability that IoT device <i>i</i> is infected by neighbor <i>n</i>
ω	Attack intention of infected IoT devices
ρ_n^i, ρ_c^i	Probability that IoT device <i>i</i> receives patches from neighbor <i>n</i> or cloud center <i>c</i>
η_n, η_c	Defense intention of deploying patches from neighbor <i>n</i> or cloud center <i>c</i>
α	Probability of requesting patches from the cloud center

a wide range of self-propagating IoT malware. This type of malware can infect increasing IoT devices by attacking known vulnerabilities.

To characterize malware propagation and patch deployment in EI-enabled IoT systems, we model the interaction among IoT devices through a state transition framework. Generally, malware propagates through communication among neighboring devices in EI-enabled IoT systems. Once an IoT device is infected, it may attempt to compromise susceptible neighboring devices and further spread the malware. Meanwhile, defense mechanisms, such as patch installation from neighboring devices or cloud servers, can help recover infected devices and suppress malware propagation. Consequently, the state of each IoT device dynamically evolves over time under the joint influence of malware infection and patch-based recovery processes. To capture this dynamic interaction, we model how neighboring devices collectively influence infection and recovery behaviors through probabilistic state transitions.

Table II summarizes the parameters that capture practical malware propagation and patch deployment characteristics in EI-enabled IoT systems. Specifically, the malware propagation parameters reflect the infection capability and attack behaviors of compromised IoT devices during network communication. For example, the infection probability parameter ϱ_n^i characterizes the probability that an IoT device becomes infected through interactions with neighboring devices, while the attack intention parameter ω models the aggressiveness of malware propagation activities. The patch deployment parameters represent the availability and effectiveness of defensive responses in IoT environments. In particular, ρ_n^i and ρ_c^i denote the probabilities that IoT devices successfully obtain patches from the neighboring device *n* or cloud server *c*, respectively, whereas η_n and η_c characterize the corresponding defense intentions for

patch deployment. In addition, the parameter α represents the probability that IoT devices request patches from cloud servers when neighboring edge nodes cannot provide timely or sufficient updates. This parameter reflects the dependence on centralized cloud-assisted patch management under resource-constrained IoT network conditions.

For any IoT device i , we consider that there are k neighbors, representing a key abstraction of the network topology. A high value of k models a well-connected subnet commonly found in industrial IoT or urban sensing networks. This implies that IoT malware has k potential pathways for propagation, leading to a complex and rapid outbreak scenario. Such a design captures the essential impact of connectivity on propagation dynamics. In typical EI-enabled IoT systems, an infected IoT device becomes malicious and attempts to attack susceptible IoT devices among its neighbors, resulting in the state transition of IoT devices from state SN to state LN , and subsequently to states IN and DN . Assume IoT device j is one of the neighbors, its infected state can be denoted as $\sigma_f^j \in \{0, 1\}$, where $\sigma_f^j = 0$ if neighbor j is uninfected, and $\sigma_f^j = 1$ if it is infected by IoT malware. Similarly, the patched state of neighbor j is denoted as $\sigma_p^j \in \{0, 1\}$, where $\sigma_p^j = 0$ if the neighbor j forgoes patch installation, and $\sigma_p^j = 1$ if it successfully installs patches. The probability ρ_n^i that IoT device i acquires patches from a neighbor n is

$$\rho_n^i = 1 - \prod_{1 \leq j \leq k} (1 - \rho_n^j \sigma_p^j), \quad (1)$$

and the probability ϱ_n^i that IoT device i is infected by a neighbor n is

$$\varrho_n^i = 1 - \prod_{1 \leq j \leq k} (1 - \varepsilon \sigma_f^j), \quad (2)$$

where ε denotes the infection rate during IoT network communication. Define $P_y^t, y \in \{SN, LN, IN, RP, DN\}$, as the probability of an IoT device being in state y at time t . Given that σ_p^j and σ_f^j correspond respectively to P_{RP}^t and P_{LN}^t , which represent the probabilities of IoT devices transitioning to states RP and LN at time t , Eqs. (1) and (2) can be reformulated as

$$\rho_n^i = 1 - \prod_{1 \leq j \leq k} (1 - \rho_n^j P_{RP}^t), \quad (3)$$

and

$$\varrho_n^i = 1 - \prod_{1 \leq j \leq k} (1 - \varepsilon P_{LN}^t). \quad (4)$$

By default, security patches are initially installed on each infected IoT device to prevent IoT malware propagation in EI-enabled IoT systems. Hence, the initial state of infected IoT devices is state RP . We define $T_{z|y}$ as the transition probability from state y to state z , where $y, z \in \{SN, LN, IN, RP, DN\}$. Specifically,

$$T_{SN|RP} = \varphi_{S|R}^i + \phi, \quad (5)$$

$$T_{LN|SN} = \varrho_n^i P_{SN}^t + \phi, \quad (6)$$

$$T_{RP|SN} = (\rho_n^i + \rho_c^i) P_{SN}^t + \phi, \quad (7)$$

$$T_{IN|LN} = \varphi_{I|L}^i + \phi, \quad (8)$$

$$T_{RP|LN} = (\rho_n^i + \rho_c^i) P_{LN}^t + \phi, \quad (9)$$

$$T_{RP|IN} = (\rho_n^i + \rho_c^i) P_{IN}^t + \phi, \quad (10)$$

$$T_{DN|LN} = \varphi_{D|L}^i + \phi, \quad (11)$$

and

$$T_{DN|IN} = \varphi_{D|I}^i + \phi, \quad (12)$$

where $\varphi_{m|l}^i, m, l \in \{S, L, I, R, D\}$, denotes the probability of IoT device i transferring from state l to state m , and ρ_c^i denotes the probability of infected IoT device i acquiring patches from cloud center c . Differences in computing power are incorporated into a systematic error ϕ , representing system failures due to resource exhaustion.

Since the time complexity increases exponentially with the number of IoT devices, determining the optimal malware patching strategy for EI-enabled IoT systems becomes significantly more challenging. Therefore, we utilize averaging techniques to simplify this problem. We consider there are a total of X IoT devices. Let $\bar{\rho}_n = \mathbb{E}(\rho_n^1, \rho_n^2, \dots, \rho_n^X)$ be the average probability that an infected IoT device receives patches from a neighbor n , and $\bar{\rho}_c = \mathbb{E}(\rho_c^1, \rho_c^2, \dots, \rho_c^X)$ be the average probability that an infected IoT device receives patches from the cloud center c . Due to limitations of central resources, infected IoT devices can request patches from the cloud center with a probability α . Besides, $\tilde{\varepsilon} = \mathbb{E}(\varepsilon^1, \varepsilon^2, \dots, \varepsilon^X)$ represents the average infection probability from a neighbor. The system of differential equations governing the dynamic state transitions of the IoT devices is given by

$$\begin{aligned} \frac{dP_{SN}^t}{dt} &= \varphi_{S|R} P_{RP}^t - k\tilde{\varepsilon} P_{SN}^t P_{LN}^t \\ &\quad - ((1-\alpha)k\bar{\rho}_n + \alpha\bar{\rho}_c) P_{SN}^t, \end{aligned} \quad (13)$$

$$\begin{aligned} \frac{dP_{LN}^t}{dt} &= k\tilde{\varepsilon} P_{SN}^t P_{LN}^t - ((1-\alpha)k\bar{\rho}_n + \alpha\bar{\rho}_c) P_{LN}^t \\ &\quad - \varphi_{I|L} P_{LN}^t - \varphi_{D|L} P_{LN}^t, \end{aligned} \quad (14)$$

$$\begin{aligned} \frac{dP_{IN}^t}{dt} &= \varphi_{I|L} P_{LN}^t - ((1-\alpha)k\bar{\rho}_n + \alpha\bar{\rho}_c) P_{IN}^t \\ &\quad - \varphi_{D|I} P_{IN}^t, \end{aligned} \quad (15)$$

$$\begin{aligned} \frac{dP_{RP}^t}{dt} &= ((1-\alpha)k\bar{\rho}_n + \alpha\bar{\rho}_c) (P_{SN}^t + P_{LN}^t + P_{IN}^t) \\ &\quad - \varphi_{S|R} P_{RP}^t, \end{aligned} \quad (16)$$

and

$$\frac{dP_{DN}^t}{dt} = \varphi_{D|L} P_{LN}^t + \varphi_{D|I} P_{IN}^t. \quad (17)$$

However, in practice, IoT malware propagation is not purely random. Infected devices selectively attack susceptible neighbors to maximize their reward. To capture this behavior, we introduce a binary attack intention $\omega \in \{0, 1\}$, where $\omega = 0$ indicates no propagation and $\omega = 1$ represents active propagation, leading to an effective infection rate of $\tilde{\varepsilon}\omega$. Similarly,

edge nodes adopt defense strategies based on a binary intention η , where $\eta_n, \eta_c \in \{0, 1\}$ denote whether patches are requested from a neighbor n or the cloud c , respectively. The corresponding recovery probability becomes $(1 - \alpha)\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c$, ranging from no recovery to maximum recovery. This binary intention modeling captures the essential strategic interaction between malware propagation and patch deployment, while the equilibrium solution reflects probabilistic action selection in realistic environments. By embedding these intentions into the state transition process, the model can effectively characterize the dynamic attacker–defender interplay. Therefore, the dynamic transition equations are updated as

$$\begin{aligned} \frac{dP_{SN}^t}{dt} = & \varphi_{S|R}P_{RP}^t - k\tilde{\omega}P_{SN}^tP_{LN}^t \\ & - ((1 - \alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c)P_{SN}^t, \end{aligned} \quad (18)$$

$$\begin{aligned} \frac{dP_{LN}^t}{dt} = & k\tilde{\omega}P_{SN}^tP_{LN}^t - ((1 - \alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c)P_{LN}^t \\ & - \varphi_{I|L}P_{LN}^t - \varphi_{D|L}P_{LN}^t, \end{aligned} \quad (19)$$

$$\begin{aligned} \frac{dP_{IN}^t}{dt} = & \varphi_{I|L}P_{LN}^t - ((1 - \alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c)P_{IN}^t \\ & - \varphi_{D|I}P_{IN}^t, \end{aligned} \quad (20)$$

$$\begin{aligned} \frac{dP_{RP}^t}{dt} = & ((1 - \alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c)(P_{SN}^t + P_{LN}^t + P_{IN}^t) \\ & - \varphi_{S|R}P_{RP}^t, \end{aligned} \quad (21)$$

and

$$\frac{dP_{DN}^t}{dt} = \varphi_{D|L}P_{LN}^t + \varphi_{D|I}P_{IN}^t. \quad (22)$$

The above equations describe the dynamic evolution of IoT device states under the joint influence of malware propagation and patch deployment. Specifically, susceptible devices can decrease due to being infected by compromised devices and being patched by neighbors or the cloud. Latent devices are mainly generated from newly infected susceptible devices. They can either evolve into actively infected devices, be directly patched and recovered, or become permanently damaged due to system failures. Infected devices arise from latent devices and can be reduced through patching or transition into damaged states. The recovery state aggregates all devices that are successfully patched from susceptible, latent, and infected states. While some recovered devices may eventually return to the susceptible state over time. Finally, the damaged state represents devices that are irreversibly affected by malware or patching failures. Overall, these equations capture the dynamic balance between malware spread and patch deployment, thereby modeling the evolving security status of the IoT system.

B. Game Model

We model the interaction between malware propagation and patching as a continuous-time differential game between IoT devices and edge nodes. IoT devices attempt to maximize infection spread by targeting vulnerable devices, while edge nodes aim to minimize system damage by deploying secure patches.

TABLE III
PAYOFF PARAMETERS

Symbol	Definition
β	Probability of data compromise caused by malware propagation
ν	Cost of data compromise caused by malware propagation
δ_n, δ_c	Cost of acquiring patches from neighbor n or cloud center c
ϵ	Reward of successful malware mitigation
ζ	Cost caused by the failure of infected IoT devices
θ	Cost of maintaining failed devices

The system evolves dynamically as IoT devices transition among various states, and both players continuously adjust their strategies based on the network condition.

In our model, malware can intelligently propagate through the network by exploiting known vulnerabilities in IoT devices. It can strategically choose naive IoT devices to infect, aiming to maximize its reward. Although the malware does not alter its propagation in real-time to directly evade a specific patching strategy, the designed attack intention reflects a high-level adaptation to the overall network state. It tends to target devices that are more susceptible to infection. Conversely, edge nodes can install authentic patches to mitigate malware propagation. For deriving the theoretical optimum, we consider they have perfect and continuous observability of the system state to dynamically adjust their defense strategies. It is worth noting that the practical algorithm we later propose does not rely on such strict rationality or full observability. It instead performs under partial observations and learns adaptive strategies directly from interactions with EI-enabled IoT environments. Furthermore, our model operates under a trusted patch distribution network, where patches are downloaded from the cloud center or neighboring devices.

Definition 1: The differential game-based malware propagation-patching (DGMPP) model for EI-enabled IoT systems is represented by a 5-tuple $DGMPP = \langle \mathcal{X}, \mathcal{A}, \mathcal{S}, \mathcal{P}, \mathcal{R} \rangle$, where:

- $\mathcal{X} = \{\text{IoT device } i, \text{ Edge node } e\}$ is the player set.
- $\mathcal{A} = \{\mathcal{A}_i^t, \mathcal{A}_e^t\}$ is the strategy set for both players at time $t \in [0, T]$, where $\mathcal{A}_i^t = \{\omega \mid \omega \in \{0, 1\}\}$ denotes the set of attack intentions for infected IoT devices at time t and $\mathcal{A}_e^t = \{(\eta_c, \eta_n) \mid \eta_c, \eta_n \in \{0, 1\}\}$ denotes the set of defense intentions for edge nodes installing patches from the cloud center c and neighbor n at time t .
- $\mathcal{S} = \{\mathcal{S}_i^t \times \mathcal{S}_e^t\}$ is the state set for both players at time $t \in [0, T]$, where $\mathcal{S}_i^t = \{S, L, I, R, D\}$ denotes the set of states for IoT devices at time t and $\mathcal{S}_e^t = \{P, N\}$ denotes the set of actions for edge nodes at time t . Thus, we have $\mathcal{S} = \{SN, LN, IN, RP, DN\}$ as the set of joint states for IoT devices under malware propagation.
- $\mathcal{P} = \{P_{SN}^t, P_{LN}^t, P_{IN}^t, P_{RP}^t, P_{DN}^t\}$ is the probability set of IoT devices being in each state at time $t \in [0, T]$.
- $\mathcal{R}(\mathcal{A}_i^t, \mathcal{A}_e^t) = \int_0^T (\mathcal{R}^t)dt + \mathcal{R}^T$ is the payoff function for both players during malware propagation and patch, where $t \in [0, T]$, $\mathcal{R}^t$ denotes the reward for intermediate states at time t , and $\mathcal{R}^T$ denotes the reward for the final state.

Table III summarizes the payoff-related parameters used in the proposed DGMPP model. We now analyze the payoff function for state transitions during malware propagation and patching in EI-enabled IoT systems. We adopt a zero-sum game by treating infected IoT devices as the attacker and edge nodes as the defender. Such a zero-sum setting is common in cybersecurity research [21], [51], where the gain of one side naturally corresponds to the loss of the other. Specifically, successful malware propagation by infected IoT devices leads to system damages including confidentiality breaches, operational disruptions, and increased maintenance costs for the entire IoT network. Edge nodes are responsible for protecting IoT system, which incurs corresponding costs when implementing defense measures. Conversely, a successful patching action by edge nodes reduces the infection probability and directly diminishes the infected IoT devices' ability to propagate malware. Although real-world infected IoT devices and associated edge nodes may pursue asymmetric objectives, the zero-sum approximation captures the strictly competitive nature of malware propagation and patching. This assumption reflects the worst-case confrontational dynamics between two players and provides a robust analytical basis for deriving equilibrium strategies. Generally, data compromise caused by malware propagation for infected IoT devices always occurs with a probability β with a cost ν , resulting in an expected penalty of $\beta\nu$. To prevent further malware propagation, edge nodes can patch and recover infected IoT devices. The cost associated with acquiring patches from neighbors and the cloud center are denoted as δ_n and δ_c , respectively. The edge node receives a reward ϵ for successfully mitigating malware propagation. Furthermore, the death of an infected IoT device, which results in a system breakdown, incurs a cost denoted by ζ . Finally, manually maintaining or replacing a dead device incurs a cost θ . Thus, the reward $\mathcal{R}^t$ for intermediate states at time t can be represented by

$$\begin{aligned} \mathcal{R}^t = & -\beta\nu(P_{LN}^t + P_{IN}^t) \\ & - (\delta_n(1-\alpha) + \delta_c\alpha)(P_{SN}^t + P_{LN}^t + P_{IN}^t) \\ & + \epsilon P_{RP}^t - \zeta P_{DN}^t, \end{aligned} \quad (23)$$

and the reward for the final state $\mathcal{R}^T$ can be represented by

$$\mathcal{R}^T = \theta P_{DN}^T. \quad (24)$$

Therefore, the payoff function $\mathcal{R}(\mathcal{A}_i^t, \mathcal{A}_e^t)$ is

$$\begin{aligned} \mathcal{R}(\mathcal{A}_i^t, \mathcal{A}_e^t) = & \int_{t=0}^T (\mathcal{R}^t) dt + \mathcal{R}^T \\ & - (\delta_n(1-\alpha) + \delta_c\alpha)(P_{SN}^t + P_{LN}^t + P_{IN}^t) \\ & + \epsilon P_{RP}^t - \zeta P_{DN}^t) dt + \theta P_{DN}^T. \end{aligned} \quad (25)$$

Definition 2: For $\forall \mathcal{A}_i^t$ and $\forall \mathcal{A}_e^t$, $(\mathcal{A}_i^{t*}, \mathcal{A}_e^{t*})$ constitutes a Nash Equilibrium for the DGMPP game if and only if

$$\mathcal{R}(\mathcal{A}_i^t, \mathcal{A}_e^{t*}) \leq \mathcal{R}(\mathcal{A}_i^{t*}, \mathcal{A}_e^{t*}) \leq \mathcal{R}(\mathcal{A}_i^{t*}, \mathcal{A}_e^t). \quad (26)$$

From the analyses above, we realize that both IoT devices and edge nodes can maximize their payoffs by adjusting their strategies until a Nash equilibrium $(\mathcal{A}_i^{t*}, \mathcal{A}_e^{t*})$ is reached. Here, $\mathcal{A}_e^{t*}$

represents the optimal patching strategy for the edge node, given the IoT device's strategy $\mathcal{A}_i^t$. To analyze the Nash equilibrium, we must first guarantee its existence in the DGMPP framework.

Theorem 1: A Nash equilibrium $(\mathcal{A}_i^{t*}, \mathcal{A}_e^{t*})$ exists for the DGMPP game in EI-enabled IoT systems.

The proof of Theorem 1 can be found in Appendix A, available online where we have considered six conditions to demonstrate $\mathcal{R}(\mathcal{A}_i^t, \mathcal{A}_e^{t*}) \leq \mathcal{R}(\mathcal{A}_i^{t*}, \mathcal{A}_e^{t*}) \leq \mathcal{R}(\mathcal{A}_i^{t*}, \mathcal{A}_e^t)$ holds for $\forall \mathcal{A}_i^t$ and $\forall \mathcal{A}_e^t$, thereby proving $(\mathcal{A}_i^{t*}, \mathcal{A}_e^{t*})$ is a Nash equilibrium for the DGMPP game.

Next, we analyze the necessary conditions for the Nash equilibrium in DGMPP utilizing Pontryagin's maximum principle (PMP). PMP provides a theoretical benchmark to give an upper bound for the open-loop control strategy, which represents the optimal malware patching strategy. We next introduce the costate variables $\lambda_{SN}^t, \lambda_{LN}^t, \lambda_{IN}^t, \lambda_{RP}^t$ and λ_{DN}^t , corresponding to the states $P_{SN}^t, P_{LN}^t, P_{IN}^t, P_{RP}^t$ and P_{DN}^t , respectively. The Hamilton function H for the DGMPP is defined as

$$\begin{aligned} H = & -\beta\nu(P_{LN}^t + P_{IN}^t) \\ & - (\delta_n(1-\alpha) + \delta_c\alpha)(P_{SN}^t + P_{LN}^t + P_{IN}^t) \\ & + \epsilon P_{RP}^t - \zeta P_{DN}^t \\ & + \lambda_{SN}^t (\varphi_{S|R} P_{RP}^t - k\tilde{\epsilon}\omega P_{SN}^t P_{LN}^t - ((1-\alpha) \\ & k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c) P_{SN}^t) \\ & + \lambda_{IN}^t (\varphi_{I|L} P_{LN}^t - ((1-\alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c) \\ & P_{IN}^t - \varphi_{D|I} P_{IN}^t) \\ & + \lambda_{RP}^t (((1-\alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c)(P_{SN}^t + P_{LN}^t \\ & + P_{IN}^t) - \varphi_{S|R} P_{RP}^t) \\ & + \lambda_{DN}^t (\varphi_{D|L} P_{LN}^t + \varphi_{D|I} P_{IN}^t), \end{aligned} \quad (27)$$

from which we can obtain

$$\begin{aligned} \frac{d\lambda_{SN}^t}{dt} = & -\frac{\partial H}{\partial P_{SN}^t} = \delta_n(1-\alpha) + \delta_c\alpha + \lambda_{SN}^t (k\tilde{\epsilon}\omega \\ & \times P_{LN}^t + (1-\alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c) + \lambda_{LN}^t k\tilde{\epsilon}\omega \\ & \times P_{LN}^t + \lambda_{RP}^t ((1-\alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c), \end{aligned} \quad (28)$$

$$\begin{aligned} \frac{d\lambda_{LN}^t}{dt} = & -\frac{\partial H}{\partial P_{LN}^t} = \beta\nu + \delta_n(1-\alpha) + \delta_c\alpha + \lambda_{SN}^t k \\ & \times \tilde{\epsilon}\omega P_{SN}^t - \lambda_{LN}^t (k\tilde{\epsilon}\omega P_{SN}^t - (1-\alpha)k\tilde{\rho}_n\eta_n \\ & - \alpha\tilde{\rho}_c\eta_c - \varphi_{I|L} - \varphi_{D|L}) - \lambda_{IN}^t \varphi_{I|L} - \lambda_{RP}^t \\ & \times ((1-\alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c) - \lambda_{DN}^t \varphi_{D|L}, \end{aligned} \quad (29)$$

$$\begin{aligned} \frac{d\lambda_{IN}^t}{dt} = & -\frac{\partial H}{\partial P_{IN}^t} = \beta\nu + \delta_n(1-\alpha) + \delta_c\alpha + \lambda_{IN}^t \\ & ((1-\alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c + \varphi_{D|I}) - \lambda_{RP}^t \\ & ((1-\alpha)k\tilde{\rho}_n\eta_n + \alpha\tilde{\rho}_c\eta_c) - \lambda_{DN}^t \varphi_{D|I}, \end{aligned} \quad (30)$$

$$\frac{d\lambda_{RP}^t}{dt} = -\frac{\partial H}{\partial P_{RP}^t} = -\epsilon - \lambda_{SN}^t \varphi_{S|R} + \lambda_{RP}^t \varphi_{S|R}, \quad (31)$$

and

$$\frac{d\lambda_{DN}^t}{dt} = -\frac{\partial H}{\partial P_{DN}^t} = \zeta, \quad (32)$$

with the terminal conditions for the costate variables given by

$$\begin{cases} \lambda_{SN}^T = 0, \\ \lambda_{LN}^T = 0, \\ \lambda_{IN}^T = 0, \\ \lambda_{RP}^T = 0, \\ \lambda_{DN}^T = \theta. \end{cases} \quad (33)$$

According to PMP, the optimal control strategies $(\mathcal{A}_i^{t*}, \mathcal{A}_e^{t*})$ must satisfy

$$(\mathcal{A}_i^{t*}, \mathcal{A}_e^{t*}) \in \arg \max_{\mathcal{A}_i^t} \min_{\mathcal{A}_e^t} H(t, \mathcal{S}, \mathcal{A}_i^t, \mathcal{A}_e^t) \quad (34)$$

and

$$(\mathcal{A}_i^{t*}, \mathcal{A}_e^{t*}) \in \arg \min_{\mathcal{A}_e^t} \max_{\mathcal{A}_i^t} H(t, \mathcal{S}, \mathcal{A}_i^t, \mathcal{A}_e^t). \quad (35)$$

The Hamilton function H in Eq. (35) can be rewritten to a linear function of the control set (ω, η_c, η_n) , represented by

$$H \triangleq \omega \tau_\omega + \eta_c \tau_c + \eta_n \tau_n + \mu, \quad (36)$$

where

$$\tau_\omega = (\lambda_{LN}^t - \lambda_{SN}^t) k \tilde{\varepsilon} P_{SN}^t P_{LN}^t, \quad (37)$$

$$\begin{aligned} \tau_c = & (\lambda_{RP}^t (P_{SN}^t + P_{LN}^t + P_{IN}^t) - \lambda_{SN}^t P_{SN}^t \\ & - \lambda_{LN}^t P_{LN}^t - \lambda_{IN}^t P_{IN}^t) \alpha \tilde{\rho}_c, \end{aligned} \quad (38)$$

$$\begin{aligned} \tau_n = & (\lambda_{RP}^t (P_{SN}^t + P_{LN}^t + P_{IN}^t) - \lambda_{SN}^t P_{SN}^t \\ & - \lambda_{LN}^t P_{LN}^t - \lambda_{IN}^t P_{IN}^t) (1 - \alpha) k \tilde{\rho}_n, \end{aligned} \quad (39)$$

and μ is the additional function. Thus, the control set (ω, η_c, η_n) can be described as

$$\omega = \begin{cases} 1, & \text{if } \tau_\omega > 0, \\ 0, & \text{if } \tau_\omega \leq 0, \end{cases} \quad (40)$$

$$\eta_c = \begin{cases} 1, & \text{if } \tau_c < 0, \\ 0, & \text{if } \tau_c \geq 0, \end{cases} \quad (41)$$

and

$$\eta_n = \begin{cases} 1, & \text{if } \tau_n < 0, \\ 0, & \text{if } \tau_n \geq 0. \end{cases} \quad (42)$$

Hereto, Eqs. (40)–(42) indicate that the Nash equilibrium of the DGMPP consists of three bang-bang controls. Specifically, when $\omega = 1$, infected IoT devices will most likely propagate IoT malware. In contrast, they will make no effort to infect the naive IoT devices if $\omega = 0$. This reduces the detection risks by remaining latent, thereby potentially achieving long-term penetration. Besides, $\eta_c = 1$ and $\eta_n = 1$ indicate that edge nodes patch aggressively using cloud or neighbor sources to ensure normal system operation. However, due to the high cost, edge nodes might not patch at all, denoted by $\eta_c = 0$ and $\eta_n = 0$. Thus, the optimal control strategies $(\mathcal{A}_i^{t*}, \mathcal{A}_e^{t*})$ are achieved if and only if the conditions in Eqs. (40)–(42) are satisfied.

IV. ALGORITHM & APPLICATION FRAMEWORK

Although PMP theoretically yields the optimal learning strategy, its application requires complete knowledge of system dynamics and precise parameter settings, which are rarely available in practical EI-enabled IoT systems. To address this limitation, we develop a learning-based approximation that can operate under partial observability and unknown environments. Specifically, we propose DGDSQ built upon the DGMPP framework. DGDSQ integrates SNNs into a standard DQN architecture, enabling the agent to learn adaptive malware mitigation strategies directly from interactions with the IoT environment. In this sense, DGDSQ can be viewed as a practical approximation of the theoretically optimal strategy derived from the game DGMPP. Unlike conventional neural networks that produce continuous outputs, SNNs communicate through discrete spike events over time. Each neuron accumulates input signals into a membrane potential and produces a spike only when a threshold is exceeded. This temporal and event-driven mechanism is well suited for modeling dynamic IoT environments, which can improve computational efficiency.

We consider an SNN has a total of L layers. Let H_l^t and V_l^t represent the membrane voltage after neural dynamics and the trigger of a spike in layer $l \in \{1, 2, \dots, L-1\}$ at time t , respectively. The dynamics of LIF neurons are given by

$$H_l^t = V_l^{t-1} + \frac{1}{\xi} (\vartheta_l S_{l-1}^t - V_l^{t-1} + V_{reset}), \quad (43)$$

and

$$V_l^t = H_l^t (1 - S_l^t) + V_{reset} S_l^t, \quad (44)$$

where ξ denotes the membrane time constant, ϑ_l denotes the learnable weight of the neurons in layer l , and V_{reset} denotes the initial membrane voltage. Besides, the output S_l^t of LIF neurons in layer $l \in \{1, 2, \dots, L-1\}$ at time t can be defined as

$$S_l^t = \Upsilon(H_l^t - V_{thre}), \quad (45)$$

where V_{thre} is the threshold voltage and the spiking function $\Upsilon(x)$ is defined as

$$\Upsilon(x) = \begin{cases} 1, & \text{if } x \geq 0, \\ 0, & \text{otherwise.} \end{cases} \quad (46)$$

The final output O_L of layer L is calculated as

$$O_L = \vartheta_L \frac{1}{T} \sum_{t=1}^T S_{L-1}^t. \quad (47)$$

Following the DQN framework, DGDSQ learns the action-value function that estimates long-term rewards. The loss function $\mathfrak{L}$ is defined as

$$\mathfrak{L} = \mathbb{E}_{(\mathcal{S}, \mathcal{A}) \sim \mathcal{D}(\cdot)} \left((y - Q(\mathcal{S}, \mathcal{A}; \vartheta))^2 \right), \quad (48)$$

where the target y is given by

$$y = \mathcal{R} + \gamma Q \left(\mathcal{S}', \arg \max_{\mathcal{A}} Q(\mathcal{S}', \mathcal{A}; \vartheta); \vartheta^- \right). \quad (49)$$

$(\mathcal{S}, \mathcal{A}) \sim \mathcal{D}(\cdot)$ denotes the minibatch sampled uniformly from the experience replay buffer, ϑ and ϑ^- represent the parameters

of the current and target networks, respectively, and γ is the discount factor. Since O_L in Eq. (47) denotes the output Q-values of DGDSQ, the loss function can be simplified as

$$\mathcal{L} = \mathbb{E}((y - O_L)^2). \quad (50)$$

Due to the non-differentiability of the spiking function $\Upsilon(x)$, standard backpropagation cannot be directly applied. We therefore employ a surrogate gradient approach to enable gradient-based optimization. The gradient for the final layer is

$$\frac{\partial \mathcal{L}}{\partial \vartheta_L} = \frac{2}{T} \mathbb{E}(O_L - y) \sum_{t=1}^T S_{L-1}^t. \quad (51)$$

For an intermediate layer $l \in \{1, 2, \dots, L-1\}$, the gradient is calculated as

$$\frac{\partial \mathcal{L}}{\partial \vartheta_l} = \sum_{t=1}^T \frac{\partial \mathcal{L}}{\partial S_l^t} \frac{\partial S_l^t}{\partial H_l^t} \frac{\partial H_l^t}{\partial \vartheta_l}. \quad (52)$$

According to Eqs. (43) and (45), the first factor $\frac{\partial \mathcal{L}}{\partial S_l^t}$ in Eq. (52) is derived as

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial S_l^t} &= \frac{\partial \mathcal{L}}{\partial S_{l+1}^t} \frac{\partial S_{l+1}^t}{\partial H_{l+1}^t} \frac{\partial H_{l+1}^t}{\partial S_l^t} \\ &= \frac{\partial \mathcal{L}}{\partial S_{l+1}^t} \frac{\partial \Upsilon(H_{l+1}^t - V_{thre})}{\partial H_{l+1}^t} \frac{\vartheta_{l+1}}{\xi}. \end{aligned} \quad (53)$$

Note that for $l = L-1$, the first factor $\frac{\partial \mathcal{L}}{\partial S_l^t}$ in Eq. (52) is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial S_l^t} &= \frac{\partial \mathcal{L}}{\partial O_L} \frac{\partial O_L}{\partial S_l^t} \\ &= \frac{2}{T} \vartheta_l \mathbb{E}(O_L - y). \end{aligned} \quad (54)$$

To address the non-differentiability of $\Upsilon(\cdot)$, we introduce a smooth surrogate function $v(\cdot)$ to approximate its gradient, enabling effective backpropagation through time. Therefore, Eq. (53) can be rewritten as

$$\frac{\partial \mathcal{L}}{\partial S_l^t} = \frac{\partial \mathcal{L}}{\partial S_{l+1}^t} \frac{\partial v(H_{l+1}^t - V_{thre})}{\partial H_{l+1}^t} \frac{\vartheta_{l+1}}{\xi}. \quad (55)$$

Besides, the third factor $\frac{\partial H_l^t}{\partial \vartheta_l}$ in Eq. (52) is derived by differentiating the neuronal dynamics equation in Eq. (43) as

$$\begin{aligned} \frac{\partial H_l^t}{\partial \vartheta_l} &= \frac{\partial H_l^t}{\partial V_l^{t-1}} \frac{\partial V_l^{t-1}}{\partial H_l^{t-1}} \frac{\partial H_l^{t-1}}{\partial \vartheta_l} + \frac{S_{l-1}^t}{\xi} \\ &= \frac{\xi - 1}{\xi} \left(1 - S_l^{t-1} + \frac{\partial v(H_l^{t-1} - V_{thre})}{\partial H_l^{t-1}} \right. \\ &\quad \left. \times (V_{thre} - H_l^{t-1}) \right) \frac{\partial H_l^{t-1}}{\partial \vartheta_l} + \frac{S_{l-1}^t}{\xi}. \end{aligned} \quad (56)$$

Finally, by recursively computing the gradient over time, we have

$$\frac{\partial H_l^t}{\partial \vartheta_l} = \begin{cases} \frac{S_{l-1}^t}{\xi}, & \text{if } t = 1, \\ \frac{\partial H_l^t}{\partial V_l^{t-1}} \frac{\partial V_l^{t-1}}{\partial H_l^{t-1}} \frac{\partial H_l^{t-1}}{\partial \vartheta_l} + \frac{S_{l-1}^t}{\xi}, & \text{if } t > 1. \end{cases} \quad (57)$$

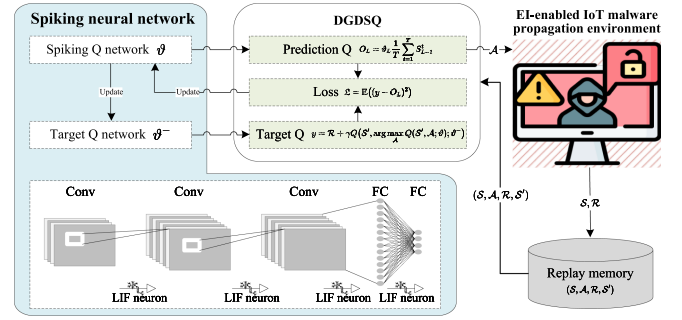


Fig. 2. DGDSQ framework for obtaining malware patching strategy in EI-enabled IoT.

We employ the arctangent function as the surrogate gradient function, which is

$$v(x) = \frac{1}{\pi} \arctan \pi x + \frac{1}{2}. \quad (58)$$

Its derivative is given by

$$v'(x) = \frac{1}{1 + (\pi x)^2}. \quad (59)$$

In summary, the architecture of the proposed DGDSQ algorithm, illustrated in Fig. 2, is based on the DQN framework. In DGDSQ, a spiking Q-network calculates the predicted Q-value according to Eq. (47) and a target Q-network calculates the target Q-value according to Eq. (49). Then, the improved loss is calculated by Eq. (50), which is then applied to update two Q-networks. The target network's parameters are periodically updated from the Q-network to enhance training stability. Furthermore, by replacing traditional DNNs with SNNs, we achieve more efficient and biologically plausible learning. The SNN architecture consists of three convolutional layers and two FC layers, each of which is connected by the LIF neuron. The complete DGDSQ algorithm is summarized in Algorithm 1.

We next analyze the computational complexity and scalability of the proposed DGDSQ. For N episodes with T steps and minibatch size B , the overall complexity is $\mathcal{O}(N \cdot T \cdot B \cdot C_{net})$, where C_{net} denotes the computational cost of a single forward and backward pass of the SNN. For an SNN with L layers and S spiking time steps, C_{net} can be approximated as $\mathcal{O}(L \cdot S \cdot d)$, where d represents the number of neurons or parameters per layer. The proposed DGDSQ adopts a compact state representation based on averaged state transition probabilities, which helps alleviate the exponential growth of the state space as the number of IoT devices increases. In addition, the learning process focuses on aggregated system dynamics rather than individual device states, which further enhances scalability in large-scale IoT environments. Moreover, the edge-centric architecture offloads most training and inference tasks to edge nodes, while IoT devices only perform lightweight operations. Communication and storage overhead can also be reduced through compact state representation and centralized replay memory management.

The proposed application framework of malware patching via a three-layer EI-enabled IoT architecture is shown in Fig. 3,

Algorithm 1: DGDSQ Algorithm to Practically Obtain the Optimal Malware Patching Strategy.

Input: Replay memory $\mathcal{D}$, total episode number N , parameter update frequency C , spiking Q-network parameter ϑ , and target network parameter ϑ^-

```

1 for episode = 1 to N do
2   Initialize environment state  $s_0$ ;
3   for  $t = 1$  to  $T$  do
4     Select action via  $\epsilon$ -greedy policy from
        $Q(s_t, a; \vartheta)$ ;
5     Execute action  $a_t$ , observe reward  $r_t$  and next
       state  $s_{t+1}$ ;
6     Store transition  $(s_t, a_t, r_t, s_{t+1})$  into  $\mathcal{D}$ ;
7     Sample minibatch from  $\mathcal{D}$ ;
8     for each transition in minibatch do
9       Forward pass through LIF neurons
         according to Eqs. (43)–(46);
10      Compute Q-value  $O_L$  according to Eq.
         (47);
11      Compute target value  $y$  according to Eq.
         (49);
12      Compute loss function  $\mathcal{L}$  according to Eq.
         (50);
13      Backward pass with surrogate gradient
         according to Eqs. (58)–(59);
14      Update parameters  $\vartheta \leftarrow \vartheta - \eta \nabla_{\vartheta} \mathcal{L}$ ;
15    end
16    if  $t \bmod C = 0$  then
17      Update target network  $\vartheta^- \leftarrow \vartheta$ ;
18    end
19  end
20 end
21 return trained results;
```

consisting of a cloud center, an edge layer, and IoT devices. In practice, our framework employs lightweight cryptographic protocols to protect all communications among IoT devices, edge nodes, and the cloud. Each device is equipped with a unique hardware-based identity to enable secure authentication during network interactions. Considering the hardware limitations of IoT devices, the proposed framework offloads the main computational tasks to edge nodes. We deploy SNN-based DRL at edge nodes, which possess stronger computational capabilities and efficiently compute optimal malware mitigation strategies. In contrast, IoT devices mainly perform lightweight operations such as state reporting and local communication with neighboring devices, which reduces the computational burden on resource-constrained devices. To ensure secure and reliable patch distribution, security patches are first retrieved and validated by the cloud center to verify their compatibility and authenticity. The validated patches are then signed, packaged, and distributed to edge nodes. The safe neighboring IoT devices also contribute to patch distribution under resource constraints. To

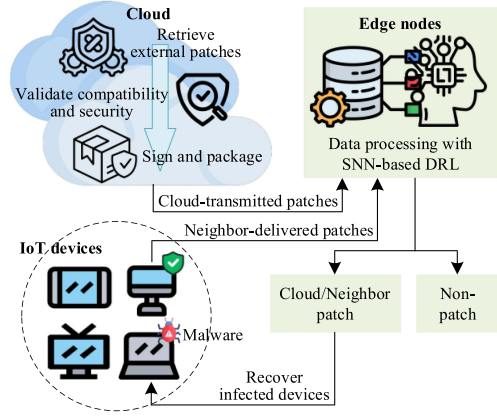


Fig. 3. The application framework of malware patching via a three-layer EI-enabled IoT architecture.

enable timely responses to malware propagation, the decision-making process for malware mitigation is performed locally at edge nodes that are close to IoT devices rather than relying solely on remote cloud servers. This reduces communication latency and allows defense strategies to be updated in near real time. Edge nodes continuously monitor device states and compute optimal malware patching strategies, while infected IoT devices propagate malware based on dynamically adjusted attack intentions. The interaction among these components constitutes a continuous dynamic process, where processed patches are distributed according to learned defense strategies.

Nevertheless, edge nodes may be compromised, and patch distribution may suffer from delays, losses, or manipulation in practical IoT environments. To account for these challenges, the proposed framework assumes semi-trusted edge nodes and secure initial device provisioning to enable tractable modeling of malware propagation and patch deployment in EI-enabled IoT systems. In practical scenarios, adversaries may attempt to inject malicious patches or distribute fake updates that compromise IoT devices. Existing security mechanisms, including cryptographic signature verification, trusted update channels, secure boot techniques, and patch validation at edge nodes or cloud servers, can be integrated with the proposed DGDSQ framework to improve the authenticity and integrity of patch updates and reduce the propagation risk of poisoned updates. Such secure patch distribution mechanisms could further enhance the reliability and robustness of patch management in EI-enabled IoT systems.

V. EXPERIMENTAL RESULTS AND EVALUATION

We implement the DGDSQ algorithm in Python to conduct experimental simulations for mitigating IoT malware propagation in EI-enabled systems. These simulations evaluate the effectiveness of DGDSQ and identify suitable parameter settings. The SNN model comprises three convolutional layers and two FC layers with learnable weights of 0.5 and an optimizer of Adam. The study adopts an analytically tractable simulation-based approach. Such epidemic-style analytical modeling has been widely utilized in IoT security research to investigate malware

dynamics under controlled and reproducible conditions [52], [53]. Furthermore, the transition parameters considered in this work are selected according to representative ranges commonly adopted in epidemic-style and cybersecurity modeling studies for networked and IoT systems [54], [55], thereby enhancing the practical relevance and interpretability of the proposed model.

In our simulation, the system dynamics are governed by transition probabilities derived from the differential equations defined in Eqs. (18)–(22). This formulation captures the behavior of large-scale IoT networks without explicitly modeling each individual device. Malware propagation is modeled as a probabilistic process, reflecting the uncertainty and heterogeneity commonly observed in real IoT environments. In IoT networks such as healthcare monitoring systems and industrial IoT infrastructures, devices are typically interconnected through local communication links and edge servers, which influence both malware propagation and patch deployment. The infection rate represents the probability of malware transmission between connected devices, which reflects network connectivity and device vulnerability. The patching probability models the effectiveness of defensive mechanisms, including security updates distributed by edge nodes or cloud services. In addition, the attack intention represents whether infected IoT devices actively attempt to propagate malware. In real IoT environments, malware may selectively launch attacks only when communication channels are available or when coordinated attack commands are received. Similarly, the defense intention represents whether edge nodes initiate patch deployment to recover infected devices. In practical IoT systems, patch distribution is often triggered by security alerts, anomaly detection results, or resource availability. The selected parameter ranges allow the simulation to represent different operational conditions.

In this paper, the learning rate, discount factor, batch size, and number of steps are set to 3×10^{-4} , 0.99, 256, and 200, respectively. These values are selected based on commonly used settings for stabilizing DRL training. Specifically, the learning rate 3×10^{-4} provides a balance between convergence speed and training stability, while the discount factor 0.99 enables the agent to consider long-term rewards in the malware mitigation process. The batch size 256 improves the stability of gradient updates during training, and 200 training steps are sufficient for the proposed DGDSQ framework to reach stable strategy convergence in our experiments. In terms of parameter sensitivity, the proposed framework is relatively robust to moderate variations of these parameters. For example, adjusting the learning rate within the range of $[1 \times 10^{-4}, 1 \times 10^{-3}]$ or the discount factor within $[0.95, 0.99]$ does not lead to significant performance degradation, but may slightly affect the convergence speed. These ranges are consistent with widely adopted settings in prior DRL literature [56], [57], where such parameter intervals have been proved to provide stable and efficient training across various tasks. However, excessively large learning rates or small discount factors may result in unstable training. Therefore, the selected parameter provides a reasonable trade-off between stability, convergence, and performance. The key experimental notations and their values are summarized in Table IV. Overall,

TABLE IV
LIST OF KEY EXPERIMENTAL NOTATIONS AND THEIR VALUES

Parameter	Notation	Value
Discount factor	γ	0.99
Training episodes	N	200
Neurons learnable weight	ϑ	0.5
Cloud patching probability	α	{0.01, 0.1, 0.4}
Infection rate	ε	{0.1, 0.3, 0.7}
Attack intention	ω	{0, 1}
Defense intention from neighbor n or cloud c	η_n, η_c	{0, 1}

the experimental design aims to capture the strategic interaction between malware propagation and patch deployment in realistic IoT environments.

A. Influence of Related Parameters

1) *Patching Probability*: This section analyzes the influence of the cloud patching probability α on the state transition rewards, the probabilities of devices being in states IN and RP , and the frequency of attack/defense intentions. The parameter α represents the effectiveness of the cloud-based repair mechanism. Cloud-based patching is often constrained by network latency and bandwidth, necessitating the complementary use of local patching from neighbors with probability $1 - \alpha$. To guarantee consistent performance, we fix the infection rate ε and adjust cloud patching probabilities α as 0.01, 0.1, and 0.4, respectively.

We first describe the rewards obtained by state transition during IoT malware propagation-patching with different patching probabilities $\alpha = 0.01$, $\alpha = 0.1$, and $\alpha = 0.4$ from the cloud center, respectively. Fig. 4(a) shows an increasing trend, meaning that IoT devices and edge nodes aim to earn maximal rewards, respectively. To be specific, edge nodes in all three cases are perpetually trained to reach the optimal malware patching strategy. It is noteworthy that DGDSQ-trained edge nodes with $\alpha = 0.1$ and $\alpha = 0.4$ arrive at 0.9 at the 12th time step, whereas those with $\alpha = 0.01$ utilize 50 time steps to reach 0.9, which demonstrates that DGDSQ-trained edge nodes with $\alpha = 0.1$ and $\alpha = 0.4$ spend less time to acquire the optimal EI-enabled IoT malware patching strategies. As a consequence, DGDSQ-trained edge nodes with $\alpha = 0.1$ and $\alpha = 0.4$ outperform those with $\alpha = 0.01$ in terms of state transition rewards.

We then describe the probability of IoT devices being in state IN during IoT malware propagation-patching with different patching probabilities $\alpha = 0.01$, $\alpha = 0.1$, and $\alpha = 0.4$, respectively. Although Fig. 4(b) has a dramatic growth in the probability of IoT devices being infected, it rapidly descends and stabilizes at 0. It means that edge nodes in all three cases are perpetually trained to recover all infected IoT devices. It is obvious that DGDSQ-trained edge nodes with $\alpha = 0.01$ cause the probability of IoT devices being infected to ascend to 0.0075 at the 12th time step, and then plunge to 0 approximately at the 60th time step. On the contrary, those with $\alpha = 0.1$ and $\alpha = 0.4$ merely make probabilities of IoT devices being infected to rise to 0.0031 and 0.0035 during the first 5 time steps, followed by

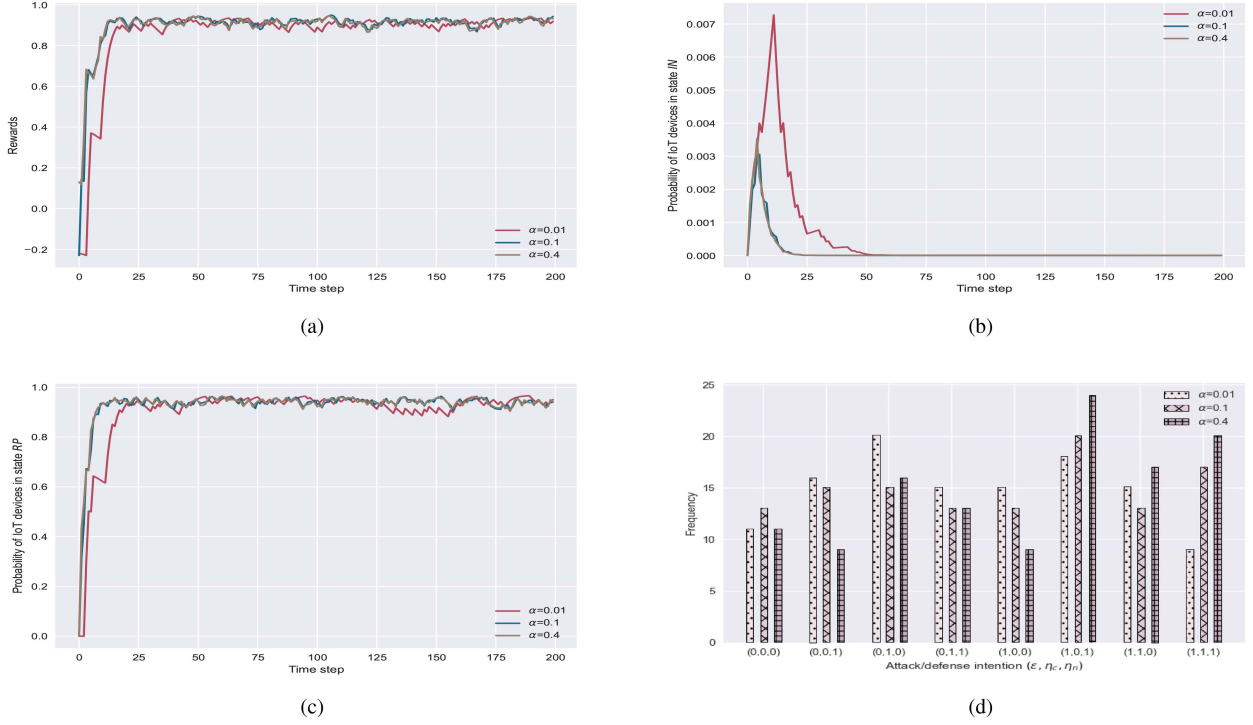


Fig. 4. Influence of cloud patching probabilities $\alpha = 0.01$, $\alpha = 0.1$, and $\alpha = 0.4$ based on DGDSQ in terms of state transition rewards, probabilities of IoT devices being in state *IN* and *RP*, as well as frequency of attack/defense intention. (a) Influence on state transition rewards. (b) Influence on probability of IoT devices being in state *IN*. (c) Influence on probability of IoT devices being in state *RP*. (d) Influence on frequency of attack/defense intention.

plummeting to 0 at the 25th time step. Thus, it takes DGDSQ-trained edge nodes with $\alpha = 0.1$ and $\alpha = 0.4$ less time to recover infected IoT devices under EI-enabled IoT malware propagation. As a consequence, DGDSQ-trained edge nodes with $\alpha = 0.1$ and $\alpha = 0.4$ outperform those with $\alpha = 0.01$ in terms of the probability of IoT devices being infected.

We next describe the probability of IoT devices being in state *RP* during IoT malware propagation-patching with different patching probabilities $\alpha = 0.01$, $\alpha = 0.1$, and $\alpha = 0.4$, respectively. Fig. 4(c) shows a rising trend, meaning that edge nodes seek to recover all infected IoT devices and guarantee the normal operation of EI-enabled IoT systems. To be specific, edge nodes in all three cases are perpetually trained to eventually patch and recover infected IoT devices. DGDSQ-trained edge nodes with $\alpha = 0.1$ and $\alpha = 0.4$ experience a sharp climb and converge to 0.95 during the first 10 time steps. However, those with $\alpha = 0.01$ cost about 50 time steps to rise to 0.95, which further shows some fluctuation around 0.9 from the 130th to the 155th time step. Such reveals that DGDSQ-trained edge nodes with $\alpha = 0.1$ and $\alpha = 0.4$ spend less time to attain the optimal EI-enabled IoT malware patching strategies. As a consequence, DGDSQ-trained edge nodes with $\alpha = 0.1$ and $\alpha = 0.4$ outperform those with $\alpha = 0.01$ in terms of the recovery probability.

We finally describe the attack/defense intention (ω, η_c, η_n) during IoT malware propagation-patching with different patching probabilities $\alpha = 0.01$, $\alpha = 0.1$, and $\alpha = 0.4$, respectively. Fig. 4(d) illustrates that DGDSQ-trained edge nodes with $\alpha = 0.4$ have a stronger defense intention in $(\omega, \eta_c, \eta_n) =$

$(1, 0, 1)$, $(\omega, \eta_c, \eta_n) = (1, 1, 0)$ and $(\omega, \eta_c, \eta_n) = (1, 1, 1)$. It means DGDSQ-trained edge nodes will request patches as much as possible from either or both of the cloud center and neighbor under the maximal malware propagation. Furthermore, compared to those with $\alpha = 0.01$ and $\alpha = 0.1$, DGDSQ-trained edge nodes with $\alpha = 0.4$ holds a relatively low frequency of $(\omega, \eta_c, \eta_n) = (0, 0, 1)$, $(\omega, \eta_c, \eta_n) = (0, 1, 0)$, $(\omega, \eta_c, \eta_n) = (0, 1, 1)$, and $(\omega, \eta_c, \eta_n) = (1, 0, 0)$, illustrating that DGDSQ-trained edge nodes with $\epsilon = 0.3$ can save more resources under the minimal attack and reduce the error rate under the maximal attack. As a consequence, DGDSQ-trained edge nodes with $\alpha = 0.4$ outperform those with $\alpha = 0.01$ and $\alpha = 0.1$ in terms of attack/defense intention.

As a corollary, cloud patching probabilities $\alpha = 0.1$ and $\alpha = 0.4$ have similar performance in respect of state transition rewards, and probabilities of IoT devices being in states *IN* and *RP*. However, the defense intention of DGDSQ-trained edge nodes with $\alpha = 0.4$ gains much more advantages over those with $\alpha = 0.01$ and $\alpha = 0.1$ especially under the maximal attack. Therefore, a cloud patching probability $\alpha = 0.4$ is identified as the optimal setting for DGDSQ.

2) *Infection Rate*: This section analyzes the influence of the infection rate ϵ on the state transition rewards, the probabilities of devices being in states *IN* and *RP*, and the frequency of attack/defense intentions. The infection rate emphasizes the overall effectiveness of attack chain, which integrates the vulnerability prevalence, network openness, and malware aggressiveness. A higher value of infection rate reflects a greater effective threat

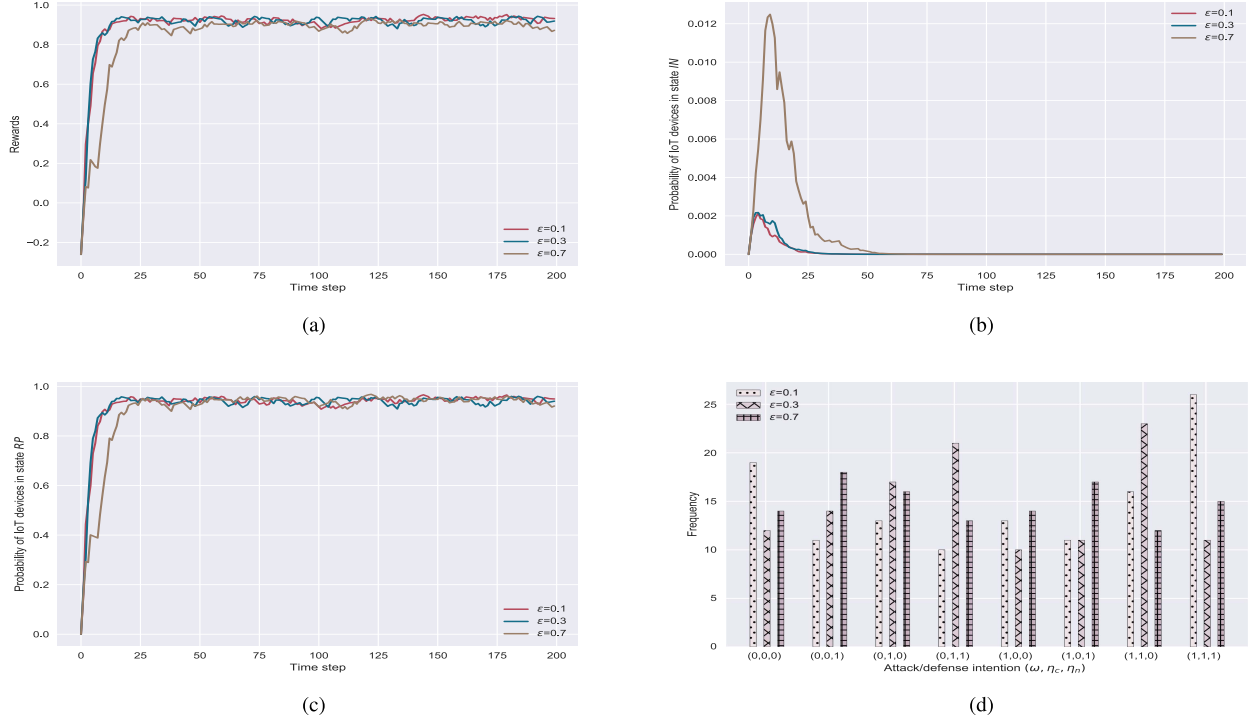


Fig. 5. Influence of infection rates $\epsilon = 0.1$, $\epsilon = 0.3$, and $\epsilon = 0.7$ based on DGDSQ in terms of state transition rewards, probabilities of IoT devices being in state *IN* and *RP*, as well as frequency of attack/defense intention. (a) Influence on state transition rewards. (b) Influence on probability of IoT devices being in state *IN*. (c) Influence on probability of IoT devices being in state *RP*. (d) Influence on frequency of attack/defense intention.

caused by IoT malware. To guarantee consistent performance, we fix the patching probability α from the cloud center as 0.4 and adjust infection rate ϵ as 0.1, 0.3, and 0.7, respectively.

We first describe the rewards obtained by state transition during IoT malware propagation-patching with different infection rates $\epsilon = 0.1$, $\epsilon = 0.3$, and $\epsilon = 0.7$, respectively. Fig. 5(a) shows an ascending trend, meaning that IoT devices and edge nodes aspire to attain maximal rewards, respectively. To be specific, edge nodes in all three cases are perpetually trained to reach the optimal IoT malware patching strategy. Noticeably, DGDSQ-trained edge nodes with $\epsilon = 0.1$ and $\epsilon = 0.3$ converge to 0.9 at the 13th time step, while those with $\epsilon = 0.7$ converge to 0.9 utilizing 70 time steps. This result indicates that DGDSQ converges faster to the optimal strategy when the infection rate ϵ is 0.1 and 0.3. As a consequence, DGDSQ-trained edge nodes with $\epsilon = 0.1$ and $\epsilon = 0.3$ outperform those with $\epsilon = 0.7$ in terms of state transition rewards.

We then describe the probability of IoT devices being in state *IN* during IoT malware propagation-patching with different infection rates $\epsilon = 0.1$, $\epsilon = 0.3$, and $\epsilon = 0.7$, respectively. Although Fig. 5(b) has a sharp growth in the probability of IoT devices being infected, it rapidly descends and stabilizes at 0, meaning that edge nodes in all three cases are perpetually trained to recover all infected IoT devices. It is apparent that DGDSQ-trained edge nodes with $\epsilon = 0.7$ make probabilities of infected IoT devices reach up to 0.012 at the 11th time step. Nevertheless, DGDSQ-trained edge nodes with $\epsilon = 0.1$ and $\epsilon = 0.3$ merely cause the probability of infected IoT devices to grow to 0.002 approximately at the 5th time step, followed

by a continuous decrease to 0 at the 25th time step. Thus, it costs DGDSQ-trained edge nodes with $\epsilon = 0.1$ and $\epsilon = 0.3$ less time to recover infected IoT devices under EI-enabled IoT malware propagation. As a consequence, DGDSQ-trained edge nodes with $\epsilon = 0.1$ and $\epsilon = 0.3$ outperform those with $\epsilon = 0.7$ in terms of the probability of IoT devices being infected.

We next describe the probability of IoT devices being in state *RP* during IoT malware propagation-patching with different infection rates $\epsilon = 0.1$, $\epsilon = 0.3$, and $\epsilon = 0.7$, respectively. It shows a climbing trend, meaning that edge nodes aim to recover all infected IoT devices and guarantee the normal operation of EI-enabled IoT systems. Specifically, edge nodes in all three cases are perpetually trained to eventually patch and recover infected IoT devices. Fig. 5(c) demonstrates that DGDSQ-trained edge nodes with $\epsilon = 0.1$ and $\epsilon = 0.3$ converge to 0.9 at the 12th time step, yet those with $\epsilon = 0.7$ converge to 0.9 utilizing around 25 time steps. It indicates it takes less time for DGDSQ-trained edge nodes with $\epsilon = 0.1$ and $\epsilon = 0.3$ to acquire the optimal EI-enabled IoT malware patching strategies. As a consequence, DGDSQ-trained edge nodes with $\epsilon = 0.1$ and $\epsilon = 0.3$ outperform those with $\epsilon = 0.7$ in terms of the recovery probability.

We finally analyze the attack/defense intention (ω, η_c, η_n) during IoT malware propagation-patching under different infection rates $\epsilon = 0.1$, $\epsilon = 0.3$, and $\epsilon = 0.7$, respectively. Fig. 5(d) shows that DGDSQ-trained edge nodes under $\epsilon = 0.3$ exhibit stronger defense intentions in high-risk scenarios, particularly for $(\omega, \eta_c, \eta_n) = (1, 1, 0)$. Meanwhile, edge nodes with $\epsilon = 0.1$ hold highest frequency in $(\omega, \eta_c, \eta_n) = (1, 1, 1)$. This means DGDSQ-trained edge nodes with both $\epsilon = 0.1$ and $\epsilon = 0.3$ will

request patches as much as possible from either or both of the cloud center and neighbor under the maximal IoT malware propagation. In addition, DGDSQ-trained edge nodes with $\varepsilon = 0.3$ reduce the frequency of (1,0,0), demonstrating fewer missed defenses under active attacks. This reflects improved resource efficiency by avoiding redundant patching. As a consequence, DGDSQ-trained edge nodes with $\varepsilon = 0.1$ and $\varepsilon = 0.3$ outperform those with $\varepsilon = 0.7$ in terms of attack/defense intention.

As a corollary, DGDSQ-trained edge nodes with infection rate $\varepsilon = 0.1$ and $\varepsilon = 0.3$ gain an advantage over those with infection rates $\varepsilon = 0.7$ in respect of state transition rewards, probabilities of IoT devices being in states *IN* and *RP*, as well as frequency of attack/defense intention. Therefore, infection rate $\varepsilon = 0.1$ and $\varepsilon = 0.3$ are identified as the optimal setting for DGDSQ. In the final algorithm comparison, we adopt $\varepsilon = 0.3$ as the representative setting. This value corresponds to a moderate infection scenario and better reflects practical IoT environments, where malware spreads but does not dominate the network. In contrast, $\varepsilon = 0.1$ represents a mild scenario with limited propagation. Using $\varepsilon = 0.3$ leads to stronger interaction between attack and defense, which makes the comparison more meaningful.

B. Algorithms Comparison

This section compares the performance utilizing double deep Q-network (DDQN), dueling double deep Q-network (D3QN), and DGDSQ algorithms on the state transition rewards, the probabilities of devices being in states *IN* and *RP*, and the frequency of attack/defense intentions. Considering that our contribution is a novel integration of differential game theory with SNN-based DRL for malware patching, the most direct and fair comparison is against other advanced DRL algorithms such as DDQN and D3QN. This comparison isolates the performance improvements attributable to our novel framework. DDQN addresses the over-estimation bias via decoupling action selection from action value evaluation, which ensures that the error in action selection does not directly affect the calculation of the target value. D3QN not only retains the strengths of DDQN, but also introduces a novel neural network structure that separates the estimation of state value function from the estimation of state-dependent action advantage functions, allowing the network to learn the value of being in a particular state independently of the action taken. Note that we include a baseline DDQN-STD. It applies DDQN under a standard epidemic model without differential game dynamics, where we remove attack/defense intentions in this model. Correspondingly, DDQN-DG and D3QN-DG represent that edge nodes attain patching strategies based on DDQN and D3QN. While applying DDQN-DG, D3QN-DG, and DGDSQ, the attack/defense intentions are constantly changing according to the chosen actions. Although other strategies like graph-based methods exist, they often require explicit global network topology for critical node identification, which is fundamentally different from our learning-based approach. Our aim is to learn an optimal adaptive strategy, whereas other methods often rely on analytical or heuristic solutions.

For this comparison, the cloud patching probability α and infection rate ε are set to 0.4 and 0.3, respectively, based on

the findings from the previous subsection. Note that transmitting patches to IoT devices all from the cloud consumes significant bandwidth and introduces latency, which is critical for time-sensitive security responses [58]. Thus, the set of cloud patching probability $\alpha = 0.4$ represents a reasonable scenario where cloud-based patching is effective but not omnipotent, necessitating the supplementary patching probability of 0.6 from neighbors. According to [59], the inherent heterogeneity of IoT devices and the strategic nature of malware propagation often result in effective infection rates below 0.5. We here set infection rate $\varepsilon = 0.3$ to ensure that the epidemic state is above the critical threshold for sustainable malware propagation [60], thereby creating a challenging environment to evaluate our patching strategies.

Fig. 6(a) illustrates the rewards obtained by state transition during IoT malware propagation-patching applying DDQN-STD, DDQN-DG, D3QN-DG, and DGDSQ approaches, respectively. DDQN-STD converges faster and achieves a higher reward because it operates under a standard epidemic model, where system dynamics are relatively simple and the reward is stable. However, this simplified setting neglects the adversarial nature of malware propagation, leading to patching strategies that may not be robust in realistic scenarios. Although the convergence of DDQN-DG becomes slower and the rewards are lower due to the uncertainty and dynamic fluctuations introduced by attacker-defender interactions, the learned patching strategies are more adaptive to real-world conditions. Besides, D3QN-DG benefits from its dueling network architecture, which separates state value and action advantage estimation, allowing it to converge more stably and achieve higher rewards. Most importantly, DGDSQ consistently outperforms all the other algorithms by combining differential game theory with SNN-based DRL, which enables the fastest convergence and obtains the highest reward. Specifically, the final rewards of DDQN-DG and D3QN-DG stabilize at approximately 0.75 and 0.9, respectively, whereas DGDSQ converges to about 0.95. Accordingly, DGDSQ achieves an improvement of approximately 27% and 6% in the converged reward compared with DDQN-DG and D3QN-DG, respectively. This superiority demonstrates that DGDSQ is better suited to capture the strategic interaction of IoT malware propagation and patching.

Fig. 6(b) illustrates the probability of IoT devices being in state *IN* during IoT malware propagation-patching applying DDQN-STD, DDQN-DG, D3QN-DG, and DGDSQ approaches, respectively. DDQN-STD-trained edge nodes cause the probability of infected IoT devices to remain consistently low. This is due to the fact that attacker/defender intentions are not explicitly expressed in the standard epidemic model, which constrains malware propagation and limits the devices entering the infected state. In contrast, DDQN-DG exhibits a more obvious infection dynamic, where the infection rate first rises and then gradually falls to zero, reflecting the influence of adopting adversarial strategies. Owing to the dueling network structure that stabilizes learning, D3QN-DG reduces the infection rate faster and more smoothly than DDQN-DG. DGDSQ achieves the best performance, as the infection probability reaches a very low value

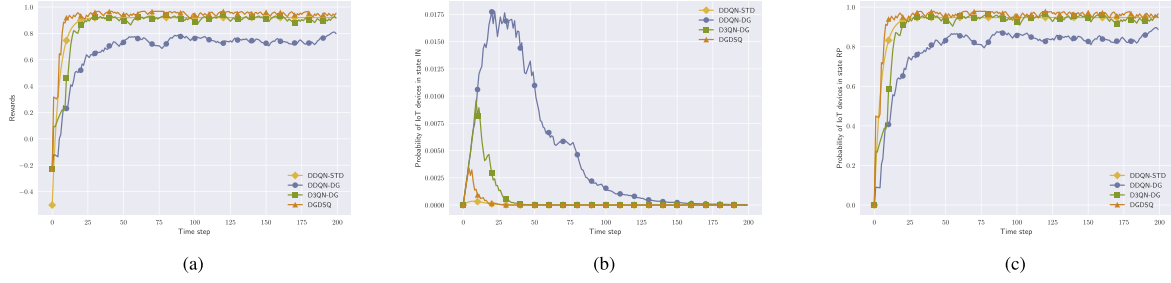


Fig. 6. Comparison among DDQN-STD, DDQN-DG, D3QN-DG, and DGDSQ in terms of state transition rewards, and probabilities of IoT devices being in states *IN* and *RP*. (a) Comparison of state transition rewards. (b) Comparison of probability of IoT devices being in state *IN*. (c) Comparison of probability of IoT devices being in state *RP*.

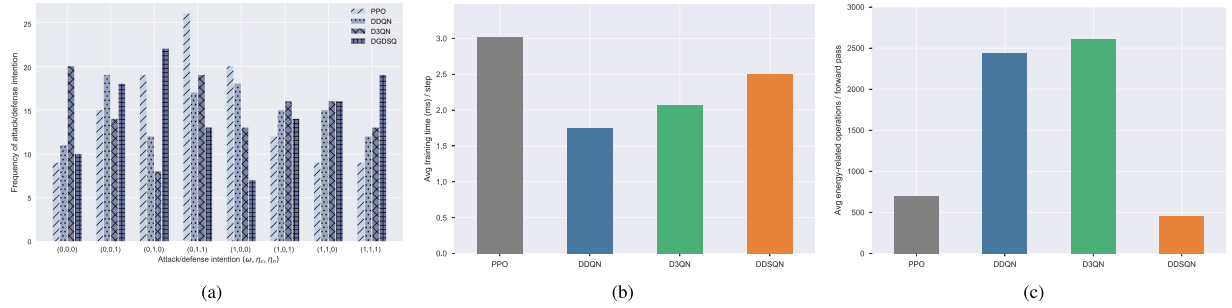


Fig. 7. Comparison among PPO, DDQN, D3QN, and DGDSQ in terms of frequency of attack/defense intention, average training time, and average energy-related operations. (a) Comparison of frequency of attack/defense intention. (b) Comparison of average training time per step. (c) Comparison of average energy-related operations per forward pass.

and quickly converges to zero. Specifically, the peak infection probabilities of DDQN-DG and D3QN-DG reach 0.0175 and 0.01, respectively, whereas DGDSQ decreases it to 0.0025. Accordingly, DGDSQ reduces the peak infection probability in state *IN* by approximately 85% and 75% compared with DDQN-DG and D3QN-DG, respectively. It demonstrates that DGDSQ-trained edge nodes are most effective in mitigating malware propagation.

Fig. 6(c) illustrates the probability of IoT devices being in state *RP* during IoT malware propagation-patching applying DDQN-STD, DDQN-DG, D3QN-DG, and DGDSQ approaches, respectively. DDQN-STD achieves a rapid increase in recovery probability and converges quickly, since the standard epidemic model lacks explicit attacker-defender strategic dynamics, making the recovery process more straightforward. In contrast, DDQN-DG converges more slowly because the differential game environment introduces adversarial interactions. D3QN-DG improves upon DDQN-DG by leveraging its dueling architecture, which stabilizes training and accelerates the increase in recovery probability. DGDSQ outperforms all other approaches, reaching the highest recovery probability and converging fastest, which maintains robustness under adversarial interactions. Specifically, the steady-state recovery probabilities of DDQN-DG and D3QN-DG stabilize at approximately 0.83 and 0.92, respectively, whereas DGDSQ reaches about 0.98. Accordingly, DGDSQ improves the steady-state recovery probability in state *RP* by approximately 18% and 7% compared with DDQN-DG and D3QN-DG, respectively. It demonstrates

that DGDSQ consistently achieves superior recovery performance and faster convergence under adversarial IoT malware propagation.

Fig. 7(a) illustrates the attack/defense intention (ω, η_c, η_n) during IoT malware propagation-patching applying PPO, DDQN, D3QN, and DGDSQ approaches, respectively. DGDSQ-trained edge nodes have a relatively strong defense intention especially in $(\omega, \eta_c, \eta_n) = (1, 0, 1)$, $(\omega, \eta_c, \eta_n) = (1, 1, 0)$ and $(\omega, \eta_c, \eta_n) = (1, 1, 1)$. It means DGDSQ-trained edge nodes will request patches as much as possible from either or both of the cloud center and neighbor under the maximal malware propagation. Moreover, the frequency of $(\omega, \eta_c, \eta_n) = (0, 1, 1)$ trained by DGDSQ is lower than those of PPO, DDQN, and D3QN, indicating that DGDSQ-trained edge nodes can conserve patch resources under the minimal malware propagation. Additionally, DGDSQ-trained edge nodes cause the least number of occurrences of $(\omega, \eta_c, \eta_n) = (1, 0, 0)$, achieving a minimal error rate. As a consequence, DGDSQ outperforms PPO, DDQN, and D3QN in terms of attack/defense intention.

Fig. 7(b) and (c) compare the computational efficiency of different DRL-based approaches in terms of average training time per step and average energy-related operations per forward pass. As shown in Fig. 7(b), PPO requires the longest average training time among all compared methods, whereas DDQN achieves the shortest training time. DDSQN introduces a moderate increase in training overhead compared with DDQN and D3QN. However, its training time remains lower than that of PPO. Fig. 7(c) further illustrates the comparison of average energy-related operations

per forward pass. Both PPO and DDSQN exhibit substantially lower operation counts than DDQN and D3QN, while DDSQN achieves the lowest operation count among all compared methods. This behavior can be attributed to the event-driven processing characteristic of SNNs, where computations are triggered only by active spikes, thereby reducing redundant computational activity through sparse neuron activations. Compared with PPO, DDSQN further reduces energy-related operations while also decreasing the average training time, indicating a more favorable tradeoff between computational overhead and operation-level computational efficiency. It is noted that the current evaluation focuses on algorithm-level computational metrics rather than direct hardware-level measurements. The reported reduction in energy-related operations is interpreted as an indicator of potential computational efficiency improvements.

As a corollary, DGDSQ-trained edge nodes gain an advantage over other approaches in respect of state transition rewards, probabilities of IoT devices being in states *IN* and *RP*, frequency of attack/defense intention, as well as energy consumption. Hence, DGDSQ can be regarded as the optimal learning algorithm.

In reality, infected IoT devices attempt to propagate malware more widely, while associated edge nodes endeavor to restrain IoT malware propagation as much as possible, thus attaining maximal benefits for both players. Hence, based on the experimental simulations above, edge nodes trained by DGDSQ can be applied in practice to patch IoT malware propagation and recover infected IoT devices in EI-enabled IoT systems. It is noticeable that setting the cloud center patching probability as 0.4 indicates that there is a probability of 0.6 that can gain patches from the neighbor, which is more effective for DGDSQ-aided edge nodes in preventing the propagation of IoT malware. Besides, setting the IoT malware infection rate as 0.3 allows IoT devices to maximize malware propagation, receiving more rewards without excessive resource consumption.

VI. CONCLUSION & FUTURE WORK

This paper has presented a novel strategy for patching IoT malware propagation in EI-enabled systems, combining differential game theory with deep spiking Q-networks. Our framework integrates AI at the network edge to enable self-learning capabilities for edge nodes. Considering the vulnerability of IoT devices to malware, we formulated a differential game-based model named DGMPP to capture the adversarial interactions between IoT devices and associated edge nodes. The DGMPP model analyzes dynamic state transitions using differential equations, and eventually determines a Nash equilibrium strategy from a theoretical perspective. The DGMPP model successfully captures the dynamic system evolution and predicts potential strategies for both players under malware propagation in EI-enabled IoT systems. Built upon the DGMPP scheme, we practically developed the DGDSQ algorithm to intelligently learn the optimal patching strategy by leveraging deep spiking Q-networks. It is noted that the proposed DGDSQ framework mainly focuses on adaptive malware patching decisions under dynamic malware propagation and attacker-defender interactions in EI-enabled IoT systems. Specifically,

DGDSQ adaptively adjusts patch deployment strategies according to changing infection states, defense conditions, and network dynamics through DRL-based learning. Our evaluation identified the patching probability and infection rate as critical parameters influencing the optimal strategy. Furthermore, the results demonstrated that DGDSQ outperforms DDQN and D3QN in mitigating malware propagation within EI-enabled IoT systems.

However, the current study relies on simulated system dynamics and several idealized assumptions to maintain analyzability. In particular, the state transition probabilities are generated through simulation, and the framework assumes accurate observability of device states by edge nodes, while IoT environments may involve incomplete observations and uncertain system dynamics. Therefore, future work will adapt the framework to real-world IoT scenarios by incorporating empirically collected data and validating the approach in controlled IoT testbeds. Such evaluations will provide stronger evidence of deployment feasibility and practical applicability. Besides, the current scalability analysis is mainly theoretical and algorithmic in nature, and comprehensive evaluations on practical large-scale IoT testbeds and heterogeneous deployment environments remain important directions for future work. Moreover, the current zero-sum formulation may also be generalized to non-zero-sum games to better capture complex attacker-defender interactions in practical cybersecurity settings. Future research may also explore cooperative defense among multiple edge nodes through multi-agent coordination mechanisms, as well as blockchain-based verification to enhance the trustworthiness of patch distribution. Furthermore, actual power consumption, inference latency, and memory footprint were not explicitly evaluated in the present implementation. A comprehensive hardware-level evaluation on practical edge platforms will be investigated in future work. In addition, the current framework does not explicitly address false-positive patching decisions, continuous adaptation to evolving malware variants, or more sophisticated adversarial scenarios such as coordinated attacks and zero-day malware. These challenges may introduce additional uncertainties in practical IoT environments and require integration with advanced malware detection and defense mechanisms. Addressing these issues remains important future work.

REFERENCES

- [1] X. Ling et al., "MalGraph: Hierarchical graph neural networks for robust windows malware detection," in *Proc. IEEE Conf. Comput. Commun., Virtual*, May 2022, pp. 1998–2007.
- [2] J. A. Alzubi, O. A. Alzubi, and I. Qiqieh, "A feature-learning-enabled malware analysis for enhanced IoT-centric cybersecurity," *Cluster Comput.*, vol. 28, Sep. 2025, Art. no. 874.
- [3] X. Pei, X. Deng, S. Tian, L. Zhang, and K. Xue, "A knowledge transfer-based semi-supervised federated learning for IoT malware detection," *IEEE Trans. Dependable Secur. Comput.*, vol. 20, no. 3, pp. 2127–2143, May/Jun. 2023.
- [4] Y. Shen, S. Shen, Q. Li, H. Zhou, Z. Wu, and Y. Qu, "Evolutionary privacy-preserving learning strategies for edge-based IoT data sharing schemes," *Digit. Commun. Netw.*, vol. 9, no. 4, pp. 906–919, Aug. 2023.
- [5] O. A. Alzubi, J. A. Alzubi, T. M. Alzubi, and A. Singh, "Quantum mayfly optimization with encoder-decoder driven LSTM networks for malware detection and classification model," *Mobile Netw. Appl.*, vol. 28, pp. 795–807, Feb. 2023.

- [6] NHS England, "NHS cyber attack: Lessons learned from Wannacry," Oct. 2018, Accessed: Sep. 30, 2025. [Online]. Available: <https://www.england.nhs.uk/long-read/case-study-wannacry-attack/>
- [7] NHS England, "Synnovis cyber attack," Jun. 2024, Accessed: Sep. 30, 2025. [Online]. Available: <https://www.england.nhs.uk/2024/06/synnovis-cyber-attack-statement-from-nhs-england/>
- [8] N. Neshenko, E. Bou-Harb, J. Crichigno, G. Kaddoum, and N. Ghani, "Demystifying IoT security: An exhaustive survey on IoT vulnerabilities and a first empirical look on internet-scale IoT exploitations," *IEEE Commun. Surv. Tuts.*, vol. 21, no. 3, pp. 2702–2733, Third Quarter 2019.
- [9] S. Jamshidi, A. Amirnia, A. Nikanjam, K. W. Nafi, F. Khomh, and S. Keivanpour, "Self-adaptive cyber defense for sustainable IoT: A DRL-based IDS optimizing security and energy efficiency," *J. Netw. Comput. Appl.*, vol. 239, Jul. 2025, Art. no. 104176.
- [10] M. Ngouen, M. A. Rahman, N. Prabakar, S. Uluagac, and L. Njilla, "Q-secure: A quantum resistant security for resource constrained IoT device encryption," in *Proc. 2023 10th Int. Conf. Internet Things: Syst. Manage. Secur.*, San Antonio, TX, USA, Nov. 2023, pp. 141–148.
- [11] G. Kuldeep and Q. Zhang, "Design prototype and security analysis of a lightweight joint compression and encryption scheme for resource-constrained IoT devices," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 165–181, Jan. 2022.
- [12] L. Zhang, J. Xu, A. Sivaraman, J. D. Lazarus, P. K. Sharma, and V. Pandi, "A two-stage differential privacy scheme for federated learning based on edge intelligence," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 6, pp. 3349–3360, Jun. 2023.
- [13] K. Wei et al., "Personalized federated learning with differential privacy and convergence guarantee," *IEEE Trans. Inf. Forensic Secur.*, vol. 18, pp. 4488–4503, 2023.
- [14] J. A. Alzubi, O. A. Alzubi, I. Qiqieh, and A. Singh, "A blended deep learning intrusion detection framework for consumable edge-centric IoMT industry," *IEEE Trans. Consum. Electron.*, vol. 70, no. 1, pp. 2049–2057, Feb. 2024.
- [15] D. Zhan, Y. Duan, Y. Hu, W. Li, S. Guo, and Z. Pan, "MalPatch: Evading DNN-based malware detection with adversarial patches," *IEEE Trans. Inf. Forensic Secur.*, vol. 19, pp. 1183–1198, Nov. 2023.
- [16] Y. Shen, C. Shepherd, C. M. Ahmed, S. Yu, and T. Li, "Comparative DQN-improved algorithms for stochastic games-based automated edge intelligence-enabled IoT malware spread-suppression strategies," *IEEE Internet Things J.*, vol. 11, no. 12, pp. 22550–22561, Jun. 2024.
- [17] A. Rezapour and W. G. Tzeng, "RL-Shield: Mitigating target link-flooding attacks using SDN and deep reinforcement learning routing algorithm," *IEEE Trans. Dependable Secur. Comput.*, vol. 19, no. 6, pp. 4052–4067, Nov./Dec. 2022.
- [18] S. Shen, C. Cai, Y. Shen, X. Wu, W. Ke, and S. Yu, "MFGD3QN: Enhancing edge intelligence defense against DDoS with mean-field games and dueling double deep Q-network," *IEEE Internet Things J.*, vol. 11, no. 13, pp. 23931–23945, Jul. 2024.
- [19] C. Chi, Y. Wang, X. Tong, M. Siddula, and Z. Cai, "Game theory in Internet of Things: A survey," *IEEE Internet Things J.*, vol. 9, no. 14, pp. 12125–12146, Jul. 2022.
- [20] S. Vajda and R. Isaacs, "Differential games. a mathematical theory with applications to warfare and pursuit, control and optimization," *Math. Gazette*, vol. 51, pp. 80–81, Feb. 1967.
- [21] H. Wu, Q. Gao, X. Tao, N. Zhang, D. Chen, and Z. Han, "Differential game approach for attack-defense strategy analysis in Internet of Things networks," *IEEE Internet Things J.*, vol. 9, no. 12, pp. 10340–10353, Jun. 2022.
- [22] Y. Xu, H. Yang, B. Jiang, and M. M. Polycarpou, "Multiplayer pursuit-evasion differential games with malicious pursuers," *IEEE Trans. Automat. Control*, vol. 67, no. 9, pp. 4939–4946, Sep. 2022.
- [23] Y. Shen, C. Shepherd, C. M. Ahmed, S. Shen, and S. Yu, "SGD3QN: Joint stochastic games and dueling double deep q-networks for defending malware propagation in edge intelligence-enabled Internet of Things," *IEEE Trans. Inf. Forensic Secur.*, vol. 19, pp. 6978–6990, Jun. 2024.
- [24] S. Yu, O. Jin, Y. Shen, G. Wu, S. Yu, and S. Shen, "Availability evaluation of industrial Internet of Things under malware propagation: An extended reliability block diagram approach based on stochastic games," *IEEE Trans. Reliab.*, vol. 74, no. 3, pp. 4253–4267, Sep. 2025.
- [25] Y. Huang and Q. Zhu, "A differential game approach to decentralized virus-resistant weight adaptation policy over complex networks," *IEEE Trans. Control Netw. Syst.*, vol. 7, no. 2, pp. 944–955, Jun. 2020.
- [26] W. Wei, J. Wang, J. Du, Z. Fang, Y. Ren, and C. L. P. Chen, "Differential game-based deep reinforcement learning in underwater target hunting task," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 462–474, Jan. 2025.
- [27] X. Ju, B. Fang, R. Yan, X. Xu, and H. Tang, "An FPGA implementation of deep spiking neural networks for low-power and fast classification," *Neural Comput.*, vol. 32, no. 1, pp. 182–204, Jan. 2020.
- [28] J. Wu et al., "Progressive tandem learning for pattern recognition with deep spiking neural networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7824–7840, Nov. 2022.
- [29] M. Pfeiffer and T. Pfeil, "Deep learning with spiking neurons: Opportunities and challenges," *Front. Neurosci.*, vol. 12, Oct. 2018, Art. no. 774.
- [30] D. Patel, H. Hazan, D. J. Saunders, H. T. Siegelmann, and R. Kozma, "Improved robustness of reinforcement learning policies upon conversion to spiking neuronal network platforms applied to atari breakout game," *Neural Netw.*, vol. 120, pp. 108–115, Dec. 2019.
- [31] M. Akter, N. Moustafa, and T. Lynar, "Edge intelligence-based privacy protection framework for IoT-based smart healthcare systems," in *Proc. IEEE Conf. Comput. Commun. Workshops*, London, U.K., May 2022, pp. 1–8.
- [32] Y. An, F. R. Yu, J. Li, J. Chen, and V. C. M. Leung, "Edge intelligence (EI)-enabled HTTP anomaly detection framework for the Internet of Things (IoT)," *IEEE Internet Things J.*, vol. 8, no. 5, pp. 3554–3566, Mar. 2021.
- [33] A. Makkar, U. Ghosh, and P. K. Sharma, "Artificial intelligence and edge computing-enabled web spam detection for next generation IoT applications," *IEEE Sens. J.*, vol. 21, no. 22, pp. 25352–25361, Nov. 2021.
- [34] C. Xu et al., "SCEL: A smart-contract driven edge intelligence framework for IoT systems," *IEEE Trans. Mob. Comput.*, vol. 23, no. 5, pp. 4453–4466, May 2024.
- [35] S. S. Wanode, M. Anand, and B. Mitra, "Optimal feature set selection for IoT device fingerprinting on edge infrastructure using machine intelligence," in *Proc. IEEE Conf. Comput. Commun. Workshops*, New York, NY, USA, May 2022, pp. 1–6.
- [36] L. X. Yang, P. Li, Y. Zhang, X. Yang, Y. Xiang, and W. Zhou, "Effective repair strategy against advanced persistent threat: A differential game approach," *IEEE Trans. Inf. Forensic Secur.*, vol. 14, no. 7, pp. 1713–1728, Jul. 2019.
- [37] L. Zhang and J. Xu, "Differential security game in heterogeneous device-to-device offloading network under epidemic risks," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 3, pp. 1852–1861, Third Quarter 2020.
- [38] Q. Gao, H. Wu, J. Zhang, Y. Zhang, N. Zhang, and X. Tao, "Multi-attacker multi-defender interaction in mMTC networks via differential game," in *Proc. IEEE/CIC Int. Conf. Commun. China*, Aug. 2020, pp. 1250–1255.
- [39] S. Ghosh and C. Konstantinou, "A bi-level differential game-based load frequency control with cyber-physical security," *IEEE Trans. Smart Grid*, vol. 15, no. 5, pp. 5151–5168, Sep. 2024.
- [40] A. C. Chapman, D. S. Leslie, A. Rogers, and N. R. Jennings, "Convergent learning algorithms for unknown reward games," *SIAM J. Control Optim.*, vol. 51, no. 4, pp. 3154–3180, 2013.
- [41] G. Liu, W. Deng, X. Xie, L. Huang, and H. Tang, "Human-level control through directly trained deep spiking Q-networks," *IEEE Trans. Cybern.*, vol. 53, no. 11, pp. 7187–7198, Nov. 2023.
- [42] K. M. Oikonomou, I. Kansizoglou, and A. Gasteratos, "A hybrid reinforcement learning approach with a spiking actor network for efficient robotic arm target reaching," *IEEE Robot. Autom. Lett.*, vol. 8, no. 5, pp. 3007–3014, May 2023.
- [43] G. Tang, N. Kumar, and K. P. Michmizos, "Reinforcement co-learning of deep and spiking neural networks for energy-efficient mapless navigation with neuromorphic hardware," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Las Vegas, NV, USA, Oct. 2020, pp. 6090–6097.
- [44] W. Tan, D. Patel, and R. Kozma, "Strategy and benchmark for converting deep q-networks to event-driven spiking neural networks," in *Proc. 35th AAAI Conf. Artif. Intell.*, Vancouver, British Columbia, Canada, Feb. 2021, pp. 9816–9824.
- [45] S. F. Chevtchenko and T. B. Ludermir, "Learning from sparse and delayed rewards with a multilayer spiking neural network," in *Proc. Int. Joint Conf. Neural Netw.*, Glasgow, U.K., Jul. 2020, pp. 1–8.
- [46] M. Saravanan, P. S. Kumar, K. Dey, S. Gaddamidi, and A. R. Kumar, "Exploring spiking neural networks in single and multi-agent RL methods," in *Proc. Int. Conf. Rebooting Comput.*, Los Alamitos, CA, USA, Nov./Dec. 2021, pp. 88–98.
- [47] O. Alrawi, C. Lever, K. Valakuzhy, K. Snow, F. Monrose, and M. Antonakakis, "The circle of life: A large-scale study of the IoT malware lifecycle," in *Proc. 30th USENIX Secur. Symp.*, Vancouver, BC, Canada, Aug. 2021, pp. 3505–3522.
- [48] X. Liu et al., "An evolutionary study of IoT malware," *IEEE Internet Things J.*, vol. 8, no. 20, pp. 15422–15440, Oct. 2021.

- [49] G. Wu, L. Xie, H. Zhang, J. Wang, S. Shen, and S. Yu, "STSIR: An individual-group game-based model for disclosing virus spread in social Internet of Things," *J. Netw. Comput. Appl.*, vol. 214, May 2023, Art. no. 103608.
- [50] A. R. Hota, J. Godbole, and P. E. Paré, "A closed-loop framework for inference, prediction, and control of SIR epidemics on networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 8, no. 3, pp. 2262–2278, Jul. 2021.
- [51] O. Tsemogne, Y. Hayel, C. Kamhoua, and G. Deugoué, "Game-theoretic modeling of cyber deception against epidemic botnets in Internet of Things," *IEEE Internet Things J.*, vol. 9, no. 4, pp. 2678–2687, Feb. 2022.
- [52] R. M. Carnier, Y. Li, Y. Fujimoto, and J. Shikata, "Exact markov chain of random propagation of malware with network-level mitigation," *IEEE Internet Things J.*, vol. 10, no. 12, pp. 10933–10947, Jun. 2023.
- [53] A. Zhaikhan, M. A. Kishk, H. ElSawy, and M. -S. Alouini, "Safeguarding the IoT from malware epidemics: A percolation theory approach," *IEEE Internet Things J.*, vol. 8, no. 7, pp. 6039–6052, Apr. 2021.
- [54] S. Gracy et al., "Modeling and analysis of a coupled SIS bi-virus model," *Automatica*, vol. 171, Jan. 2025, Art. no. 111937.
- [55] Z. Masood, R. Samar, and M. A. Z. Raja, "Design of a mathematical model for the stuxnet virus in a network of critical control infrastructure," *Comput. Secur.*, vol. 87, Nov. 2019, Art. no. 101565.
- [56] H. Yang, Z. Xiong, J. Zhao, D. Niyato, L. Xiao, and Q. Wu, "Deep reinforcement learning-based intelligent reflecting surface for secure wireless communications," *IEEE Trans. Wirel. Commun.*, vol. 20, no. 1, pp. 375–388, Jan. 2021.
- [57] A. M. Y. Cao and R. Naha, "A real-time defense framework using PPPO in deep reinforcement learning for CyberBattleSim," in *Proc. 24th Int. Conf. Trust Secur. Privacy Comput. Commun.*, Guiyang, China, Nov. 2025, pp. 3024–3031.
- [58] Z. Sun et al., "Cloud-edge collaboration in industrial Internet of Things: A joint offloading scheme based on resource prediction," *IEEE Internet Things J.*, vol. 9, no. 18, pp. 17014–17025, Sep. 2022.
- [59] G. Wu, Y. Zhang, H. Zhang, S. Yu, S. Yu, and S. Shen, "SIHQR model with time delay for worm spread analysis in IIoT-enabled PLC network," *Ad Hoc Netw.*, vol. 160, Jul. 2024, Art. no. 103504.
- [60] R. Pastor-Satorras, C. Castellano, P. Van Mieghem, and A. Vespignani, "Epidemic processes in complex networks," *Rev. Mod. Phys.*, vol. 87, no. 3, pp. 925–979, Aug. 2015.



Yizhou Shen received the BSc (Hons.) degree in computer science from Cardiff University, Cardiff, U.K., in 2023. She is currently working toward the PhD degree in computer science with Newcastle University, Newcastle upon Tyne, U.K. Her research interests include cyber security, game theory, deep reinforcement learning, edge intelligence, and the Internet of Things.



Carlton Shepherd received the BS degree in computer science from Newcastle University, U.K., in 2013, and the PhD degree in information security from the Royal Holloway, University of London, U.K., in 2019. He is currently an assistant professor in computer science with Durham University, U.K.. Before this role, he was a lecturer in computer science with Newcastle University and a senior research fellow with the Information Security Group (ISG), Royal Holloway, University of London. His research interests include centre around the security of trusted execution environments (TEEs) and their applications, secure CPU design, embedded systems, applied cryptography, and hardware security.



Chuadhry Mujeeb Ahmed received the PhD degree in information systems technology and design from the Singapore University of Technology and Design (SUTD), Singapore, in 2019, and the MSc degree in electrical engineering and computer science from Seoul National University South Korea, in 2012. He is currently a senior lecturer in computer science with Newcastle University. During the Ph.D., he worked on Cyber-Physical Systems Security, iTrust Labs and Testbeds, at SUTD. His research interests include the security and privacy of cyber-physical systems, the Internet of Things, communication systems, and critical infrastructures.



Shigen Shen (Senior Member, IEEE) received the PhD degree in pattern recognition and intelligent systems from Donghua University, Shanghai, China, in 2013. He is a professor with the School of Information Engineering, Huzhou Normal University, Huzhou, China. He has published more than 100 respected technical papers. His current research interests include Internet of Things, cyber security, edge computing, and game theory.



Shui Yu (Fellow, IEEE) received the PhD degree in computer science from Deakin University, Melbourne, VIC, Australia, in 2004. He is a professor with the School of Computer Science, University of Technology Sydney, Sydney, NSW, Australia. He has published three monographs and edited two books, more than 600 technical papers. His H-index is 75. His research interest includes Big Data, security and privacy, networking, and mathematical modeling. He is currently serving for number of prestigious editorial boards, including *IEEE Communications Surveys and Tutorials* (Area Editor), *IEEE Transactions on Computational Social Systems*, and *IEEE Internet of Things Journal*.